



Review article

Explaining explainability: A comprehensive survey on explainable artificial intelligence and relevant industry applications

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ABSTRACT

As artificial intelligence (AI) systems become increasingly integral to decision-making across industries, particularly in high-stakes domains such as healthcare, finance, and autonomous vehicles, there is a pressing need for explainable models that enhance transparency, interpretability, and accountability. The field of Explainable Artificial Intelligence (xAI) has emerged to address the opacity of complex “black-box” models, yet existing literature lacks a structured overview that organizes xAI techniques according to specific machine learning paradigms. This survey addresses the gap in existing literature by investigating how xAI techniques can be systematically organized and applied across different machine learning paradigms, including Supervised Learning, Unsupervised Learning, Reinforcement Learning, Computer Vision, and Generative AI models. The research problem encompasses understanding the relationships between xAI methods and their applicability to various AI model types, as well as identifying key challenges and limitations in current xAI implementations. This paper aims to provide a comprehensive survey that offers a structured overview of xAI techniques tailored to specific machine learning paradigms, serving as both a primer and a reference for researchers, AI developers, and industry professionals seeking to implement explainability in their AI-driven systems.

This survey synthesizes and organizes the extensive body of existing xAI literature according to machine learning paradigms, examining both ante-hoc and post-hoc explainability approaches while exploring their applications across diverse industries including healthcare, retail, finance, and manufacturing. Through this comprehensive review, several critical patterns emerge from the literature: a significant gap exists regarding generalizable metrics to compare and evaluate xAI techniques, making it difficult to assess which methods outperform others; fundamental trade-offs persist between model interpretability and accuracy, with transparent models still being preferred when explainability heavily outweighs accuracy requirements; the field lacks standardized evaluation frameworks despite recent attempts to develop comprehensive metrics; and xAI adoption faces challenges related to generating explanations digestible for non-technical stakeholders. The synthesis reveals that xAI techniques can be effectively organized by machine learning paradigm, providing practitioners with a structured approach to selecting appropriate explanation methods based on their specific model types.

This comprehensive survey provides a detailed taxonomy of xAI methods organized by machine learning paradigm, addressing a gap in current survey literature. By linking xAI techniques to specific ML paradigms and exploring their applications across industries, this work supports the enhancement of trustworthiness and effectiveness of AI systems. The identification of key challenges, particularly the lack of standardized evaluation metrics and the interpretability-accuracy trade-off, highlights critical areas for future research and development in the field of explainable AI.

1. Introduction

As artificial intelligence (AI) systems become increasingly sophisticated and pervasive across critical domains—from healthcare diagnostics and financial decision-making to autonomous vehicles and legal systems—the demand for transparency and accountability in AI

decision-making has reached unprecedented levels. The deployment of complex machine learning models in high-stakes applications has created an urgent need for methods that can explain how these systems arrive at their predictions and recommendations. This need is driven not only by regulatory requirements and ethical considerations but

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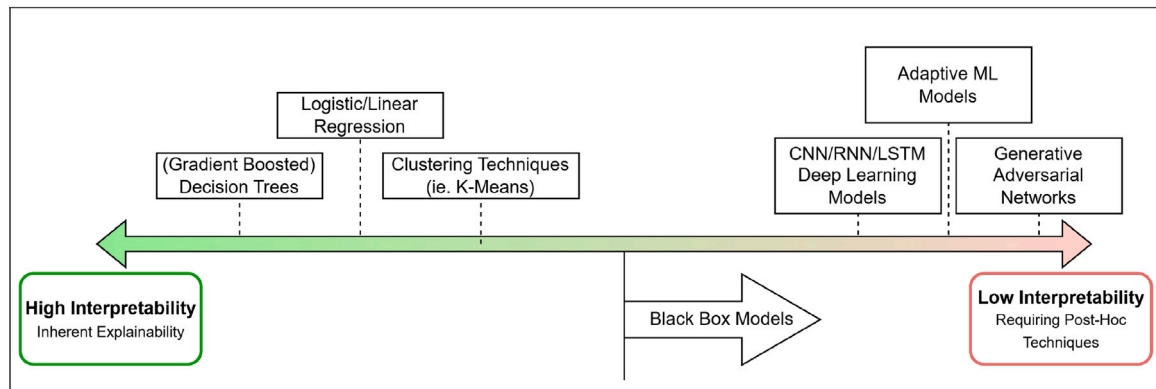
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Table 1

Table outlining commonly used “white box” algorithms and an explanation of their interpretability.

Model	Description	Interpretability explanation
Linear/Logistic Regression	Logistic regression is a classification model that predicts a binary dependent variable. When the dependent variable is continuous, the regression can be considered linear.	Goodness of fit statistics (such as chi-squared) can inform users how well the model fits the data, and algorithmic/graphing techniques can identify where the developed regression function misaligns with the desired input/output relationship. Visualization becomes impaired when the decision tree complexity increases.
K-Nearest Neighbors	KNN is a supervised classification algorithm that uses labeled training data to classify new instances based on the k nearest neighbors in the feature space.	KNN provides interpretability by showing which training examples (neighbors) influence each prediction. Users can examine the k nearest neighbors used for classification, understand the similarity metrics that determine neighbor selection, and identify which features are most important in the distance calculation. This transparency allows users to understand why a particular instance is classified in a certain way based on the most similar training examples.
General Additive Models	GAMs are linear models where the output is determined by the aggregation of a number of unknown functions defined for the predictor variables. The GAM function will infer these smooth functions, whose aggregate predicts the variable.	Such models are by definition mapping each input variable's trajectory to get to the output. As a result, it can easily be identified using statistical and visualization methods, how the model will react to certain inputs, and obtain certain outputs.

**Fig. 1.** Levels of interpretability for varied ML approaches and algorithms.

also by the practical necessity of building trust among stakeholders, including end-users, domain experts, regulators, and the broader public. The field of Explainable Artificial Intelligence (xAI) has emerged as a critical research area dedicated to addressing this challenge, developing techniques that can illuminate the decision-making processes of AI systems that would otherwise remain opaque “black boxes.”

Cognitive systems are dynamic systems that can perceive their environment and execute intelligent actions within it to gain additional information. Cognition encapsulates four key components: the perception-action cycle, attention, memory and intelligence (Horiushkina et al., 2018). Through the use of the perception-action cycle, a system can retain any obtained information using attention and memory, and learn from its environment to dictate future actions. The capacity of choosing future actions is permitted through the system's use of intelligence. Intelligence can come in many forms, but ultimately involves some sort of decision-making process based on the information obtained from the environment and is dependent on the presence of a feedback system within the process. The feedback system allows the overall system to learn from environments encountered due to previous actions taken and informs future actions as a result. Many algorithms have been used to help make these decisions, but as technology continues to improve, so does the complexity of these decision-making models.

Early intelligence models used basic statistical algorithms such as regression, tree-based decision logic or general additive models. These provided cognitive systems with simple learning-based intelligence, and a degree of interpretability on their own. These types of simple, yet explainable models have been labeled “white-box” or “transparent”

models (Arrieta et al., 2020). These algorithms can be used for a myriad of problems such as forecasting, classification, risk analysis and many others. Following these, more computationally intensive machine learning techniques using neural networks were developed, and these deep learning algorithms provided more accurate solutions to similar, and new, use cases. Today, the less accurate, transparent approaches are still commonly used, largely due to their high interpretability. For instance, simple decision tree models can easily be plotted as a tree-type graph, which showcases exactly how the model is making decisions. However, more complex tree-based approaches, such as gradient-boosted tree algorithms, present a different scenario: while individual trees within the ensemble can be visualized, the ensemble itself consists of many trees, making global interpretability non-trivial. For gradient-boosted models, interpretability typically relies on post-hoc tools such as feature importance scores, SHAP values, partial dependence plots, or surrogate models to understand the overall model behavior, rather than direct visualization of the entire ensemble (Saberian et al., 2019). This interpretability, whether inherent or achieved through post-hoc methods, is highly advantageous, and often required, for many businesses. Some common “white-box” methods are outlined in Table 1, along with their explainable features. Recently developed complex intelligent models, often score significantly higher in accuracy than traditional transparent models and can be used for a wider range of applications (Bengio et al., 2021). Unfortunately, many of these solutions are models that are not transparently explainable by design, and therefore sacrifice explainability for accuracy. These inherently non-transparent models will be referred to as “black-box” models in this paper and require a type of interpretation denoted as

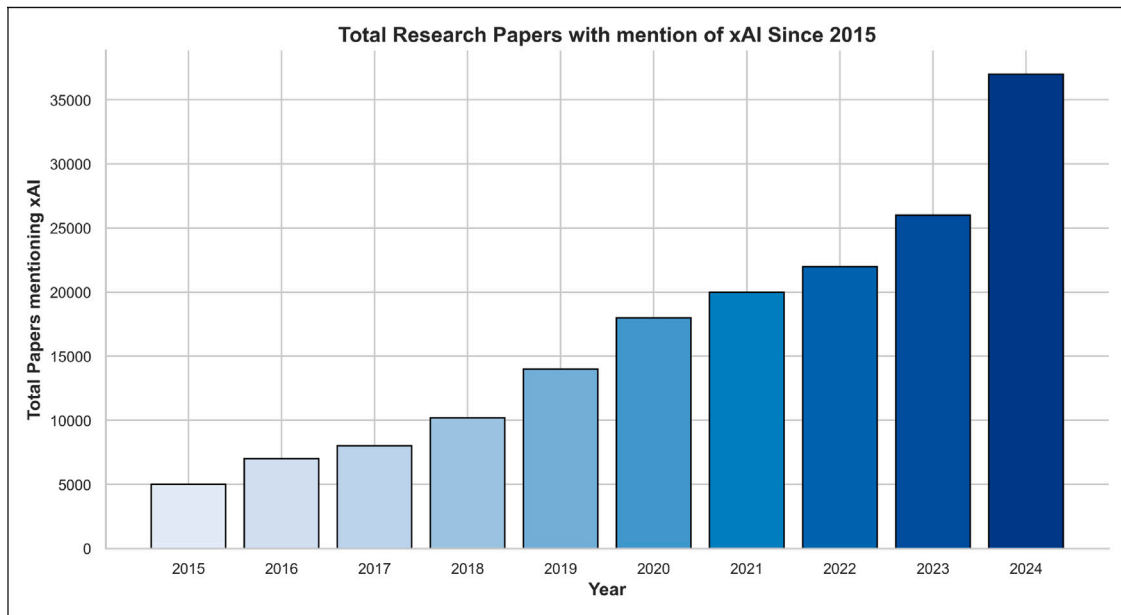


Fig. 2. Research publications referencing explainable artificial intelligence (xAI) since 2015 based on arXiv archives. Data collected using arXiv API (<http://export.arxiv.org/api/query>) with search query “xAI” across years 2015–2024. The search was performed by iterating through each month of each year (using submittedDate ranges) to bypass the 1000-paper API limit, with batch sizes of 500 papers per request, sorted by submittedDate in descending order. Inclusion criteria: papers containing “xAI” in title, abstract, or metadata as returned by the arXiv API search. Note: The search may include papers that mention xAI in passing rather than as a primary focus, and the counts represent total mentions rather than papers exclusively focused on xAI.

“post-hoc” (Arrieta et al., 2020). This introduces a spectrum of explainability for various models, which may require varying explainable techniques. A sample of this explainability spectrum is shown in Fig. 1. As a result, businesses often refuse to use them, since they cannot interpret why these algorithms are making the decisions that they are Robison (2022). Specifically, in fields with high risk, such as the medical field (Holzinger et al., 2022), explainability is of the utmost importance, and it can ultimately become more important than the accuracy of the model itself.

This problem sparked a new field of research called Explainable AI (or xAI) (Samek et al., 2017). This field explores how to apply techniques to explain the decision-making processes of non-transparent models, and as a result, enable human experts to understand why model decisions are being made. Ultimately, the goal of xAI is to make many of these powerful “black-box” algorithms more generally accepted in practice. xAI methods are divided into those that attempt to convey the inner workings of a model (local explainability), and those that attempt to explain global model functionality without identifying the mechanisms by which the models specifically make their decisions (global explainability) (Holzinger et al., 2020; Lipton, 2016). This gradual increase in interest can be seen in Fig. 2, with a constantly growing community of researchers taking an active interest in it, although this exponential growth means there is often novel and overlapping contributions in the space. This paper intends to consolidate many of those ideas, and recommend several approaches to real world scenarios, by explicitly linking xAI techniques to industry scenarios.

Before proceeding, it is crucial to establish clear distinctions between two fundamental concepts that are often used interchangeably but represent distinct aspects of understanding AI systems: *interpretability* and *explainability*. While these terms are frequently conflated in the literature, this paper follows the definitions established by Das and Rad (2020) and Merisha et al. (2024), wherein they are regarded as interconnected yet separate ideas.

Interpretability refers to the ability to comprehend exactly how a model generates its output by directly examining the model’s internal structure and parameters (Craig & Wisbey, 2024; Gilpin et al., 2018). This concept applies primarily to “white-box” models that are

inherently transparent by design. For example, a decision tree is interpretable because one can trace the exact path from input features through the tree’s nodes to the final prediction, understanding precisely which conditions led to the decision. Similarly, linear regression models are interpretable because the coefficients directly show how each input feature contributes to the output. However, interpretability is limited to models whose internal mechanisms can be directly understood, and such models often face the interpretability-accuracy trade-off when compared to more complex deep learning models.

In contrast, *Explainability* involves conveying the behavior of a machine learning model in understandable, human terms, without necessarily requiring direct access to or understanding of the model’s internal structure (Craig & Wisbey, 2024; Gilpin et al., 2018). Explainability can be applied to both interpretable (white-box) and non-interpretable (black-box) models. For instance, while a deep neural network is not interpretable (one cannot easily understand how its millions of parameters produce a prediction), it can be made explainable through post-hoc techniques such as LIME or SHAP, which generate human-understandable explanations of the model’s predictions. These techniques might indicate which input features were most important for a particular prediction, or provide a simplified local model that approximates the complex model’s behavior in a specific region of the input space. The key distinction is that interpretability requires direct model transparency, while explainability focuses on generating understandable descriptions of model behavior, regardless of the model’s internal complexity.

This distinction is critical for understanding the scope and applicability of different xAI techniques. Interpretable models (e.g., decision trees, linear models, rule-based systems) provide inherent interpretability but may sacrifice accuracy. Explainable techniques (e.g., LIME, SHAP, attention visualizations) enable explainability for complex models, allowing practitioners to use high-accuracy black-box models while still providing explanations to stakeholders. A more detailed discussion of these concepts and their applications is provided in Section 2.1.

The increasing use of AI in decision-making processes across various industries has led to the urgent need for more transparent, interpretable, and explainable models. This paper aims to bridge the gap

between complex machine learning models and their application in real-world cognitive systems by focusing on Explainable AI (xAI). While there is a growing body of work on xAI, largely focusing on generalized techniques built for various contexts and model applications, this paper distinguishes itself by offering a structured overview of the terminologies and techniques relative to distinct AI techniques. Specifically, this paper outlines commonly used approaches of AI in the context of Supervised Learning ML techniques, Unsupervised Learning ML techniques, Reinforcement Learning ML techniques, Computer Vision ML models and Generative AI models. This paper also concludes with a comprehensive analysis of various recent applications of xAI across a wide range of domains, ensuring a comprehensive understanding of both theoretical foundations and practical applications, making it highly relevant for professionals and researchers seeking to implement explainability in their AI-driven systems.

1.1. Objectives and key contributions

This survey paper aims to provide a structured overview of xAI techniques organized by machine learning paradigms, addressing a gap in existing survey literature where techniques are typically presented in a generalized manner rather than being linked to specific ML paradigms. Additionally, it comprehensively covers terminology and foundational concepts, clearly distinguishing between interpretability and explainability, as well as ante-hoc and post-hoc approaches. The survey explores industry applications of xAI across diverse domains to demonstrate practical relevance and real-world impact, while also identifying key challenges, limitations, and future research directions in the field of xAI.

The key contributions of this work include a comprehensive taxonomy of xAI methods organized by machine learning paradigm (Supervised Learning, Unsupervised Learning, Reinforcement Learning, Computer Vision, and Generative AI), which addresses a gap in current survey literature by providing a structured approach to selecting appropriate explanation methods based on the underlying ML paradigm. The survey also provides coverage of both ante-hoc and post-hoc explainability approaches, offering readers a complete understanding of the xAI landscape. Through detailed discussion of industry applications across healthcare, retail, finance, and manufacturing sectors, the work illustrates the practical significance of xAI in high-stakes decision-making environments. Furthermore, the identification of critical research gaps, particularly the lack of standardized evaluation metrics and the interpretability-accuracy trade-off, highlights areas requiring future research and development. Finally, the survey provides a structured framework that serves as both a primer for newcomers to the field and a comprehensive reference for researchers and practitioners seeking to implement explainability in their AI-driven systems.

1.2. Direction and scope of the study

This survey adopts a systematic approach to reviewing and synthesizing existing literature on Explainable AI, with a specific focus on organizing techniques according to their application across different machine learning paradigms. The scope of this work encompasses foundational terminology and definitions, with particular emphasis on distinguishing between interpretability and explainability. It includes detailed exploration of common xAI techniques (e.g., LIME, SHAP, PDP, ICE, Anchors) and their applications, as well as paradigm-specific xAI approaches for supervised learning, unsupervised learning, reinforcement learning, computer vision, and generative AI models. The survey also examines real-world industry applications demonstrating practical implementation and impact, and addresses challenges, limitations, and future directions in the field.

The organizational structure of this survey is designed to guide readers from foundational concepts through specific techniques and applications, culminating in a discussion of challenges and future

opportunities. This paper is intended for a diverse audience including: researchers in artificial intelligence and machine learning seeking a comprehensive overview of xAI techniques; AI developers and engineers implementing explainability in production systems; industry professionals in high-stakes domains (healthcare, finance, manufacturing) requiring transparent AI decision-making; and graduate students and newcomers to the field seeking a structured introduction to explainable AI. By providing a paradigm-specific organization of xAI techniques, this survey enables readers to quickly identify relevant explanation methods based on their specific ML model type, thereby supporting more effective implementation of explainability in practice.

1.3. Survey methodology

This survey employs a systematic review methodology to synthesize existing literature on Explainable AI. The methodology begins with comprehensive literature search across major databases and repositories (e.g., arXiv, IEEE Xplore, ACM Digital Library, Google Scholar) using keywords related to explainable AI, interpretable machine learning, and xAI techniques. This is followed by identification and analysis of key xAI techniques and their applications across different machine learning paradigms. The techniques are then organized according to their underlying ML paradigm (supervised, unsupervised, reinforcement learning, computer vision, generative AI) to provide a structured taxonomy. The methodology also involves synthesis of terminology, definitions, and foundational concepts from authoritative sources in the field, examination of industry applications and case studies to demonstrate practical relevance, and identification of challenges, limitations, and research gaps through critical analysis of the literature.

The “results” of this survey paper are the synthesis, taxonomy, and insights derived from reviewing the literature, which are presented throughout the paper. These include: the structured taxonomy of xAI techniques organized by ML paradigm; identification of key findings such as the gap in generalizable evaluation metrics and trade-offs between interpretability and accuracy; comprehensive coverage of both theoretical foundations and practical applications; and identification of critical areas for future research. This systematic approach ensures that the survey provides a comprehensive, well-organized, and accessible overview of the explainable AI landscape.

1.4. Scope and alternative approaches

In conducting this survey, we have made deliberate scope decisions to maintain focus on core xAI methodologies for explaining AI/ML model decisions. While the field of explainable AI intersects with various domains, we have prioritized techniques that directly address the explanation of model predictions and decision-making processes.

It is worth noting that during the review process, several alternative approaches were suggested, including fuzzy control systems for partial differential equations (PDEs) (Song et al., 2025) and event-triggered control for Markov jumping systems (Song et al., 2024). While these papers demonstrate valuable applications of AI/ML techniques in control systems, they are primarily focused on control theory applications rather than core xAI methodologies for explaining AI/ML model decisions. These approaches address control system design and performance optimization rather than the fundamental question of how to explain why an AI model makes a particular prediction or decision.

Readers interested in explainability within control systems may find these approaches valuable for understanding how AI/ML techniques can be applied in control contexts, but they do not contribute directly to the understanding of explainability techniques, evaluation methods, or xAI applications that are the focus of this survey. Our survey maintains a focused scope on methods and techniques for explaining AI model decisions across different ML paradigms, while acknowledging that alternative approaches and domain-specific applications exist that may complement the core xAI methodologies discussed herein.

1.5. Research gaps and motivation

Despite the growing importance of xAI, several critical gaps persist in the current literature that limit the field's advancement and practical adoption. First, existing survey papers typically present xAI techniques in a generalized manner without systematically organizing them according to their application across different machine learning paradigms. This organizational gap makes it difficult for practitioners to identify appropriate explanation methods for their specific model types, whether they are working with supervised learning models, unsupervised clustering algorithms, reinforcement learning agents, computer vision systems, or generative AI models. Second, the field lacks standardized evaluation metrics and frameworks for comparing xAI techniques, making it challenging to assess which methods are most effective for specific applications or to measure improvements in explanation quality. Third, there is limited understanding of the trade-offs between explanation quality, computational efficiency, and model performance, particularly for large-scale and high-dimensional applications. Fourth, the relationship between different explanation approaches (ante-hoc versus post-hoc, local versus global) and their appropriate use cases remains under-explored in a systematic manner. Finally, while numerous xAI techniques have been developed, there is insufficient guidance on how to select and combine methods to address the diverse needs of different stakeholders and application domains.

These gaps have significant implications for both research and practice. From a research perspective, the lack of systematic organization hinders the identification of areas where new techniques are most needed and makes it difficult to build upon existing work effectively. From a practical standpoint, the absence of clear guidance on technique selection and evaluation limits the adoption of xAI in real-world applications, particularly in regulated industries where explainability is not merely desirable but legally required. This survey addresses these gaps by providing a structured, paradigm-specific organization of xAI techniques, identifying critical research challenges, and offering practical guidance for technique selection and implementation.

1.6. Implications and impacts

The implications of this survey extend across multiple dimensions of the xAI field. For researchers, this work provides a structured framework that can guide future research directions by clearly identifying gaps in current knowledge and highlighting areas where new techniques are most needed. The paradigm-specific organization enables researchers to understand how xAI techniques relate to different machine learning approaches, facilitating more targeted and effective research efforts. Additionally, by synthesizing evaluation practices and identifying limitations of current methods, this survey can inform the development of new evaluation frameworks and standardized metrics.

For practitioners and industry professionals, the survey's impact lies in its practical utility for selecting and implementing xAI techniques in real-world applications. The paradigm-specific organization enables AI developers to quickly identify relevant explanation methods based on their specific model types, reducing the time and effort required to implement explainability in production systems. The comprehensive coverage of industry applications demonstrates the practical significance of xAI across diverse domains, providing concrete examples that can guide implementation decisions. Furthermore, the identification of scalability considerations and trade-offs helps practitioners make informed decisions about which xAI approaches are most suitable for their specific computational and accuracy requirements.

The broader impact of this work on the field of AI and society is substantial. By facilitating the adoption of explainable AI systems, this survey contributes to the development of more trustworthy and accountable AI technologies. In high-stakes domains such as healthcare, finance, and autonomous systems, improved explainability can enhance safety, fairness, and regulatory compliance. The survey's emphasis on

addressing the needs of non-technical stakeholders recognizes that effective explainability must serve diverse audiences, from technical developers to end-users with limited AI expertise. Ultimately, by making xAI techniques more accessible and systematically organized, this work supports the broader goal of developing AI systems that are not only powerful and accurate but also transparent, trustworthy, and aligned with human values and regulatory requirements.

This comprehensive survey paper is organized as follows. Section 2 offers an overview of common terminology, distinguishing between interpretability (white box) and explainability (black box), and introduces both ante-hoc and post-hoc techniques. In Section 3, we delve into a detailed introduction of common xAI techniques, such as LIME, Shapley Values, and PDP, among others. Sections 4 to 8 focus on the application of explainability in different learning methods, including supervised (Section 4), unsupervised (Section 5), and reinforcement (Section 6) learning, as well as specific techniques for computer vision (Section 7) and GANs (Section 8). Section 9 discusses the various industries where xAI is being applied, including healthcare, retail, finance, and manufacturing. Finally, Section 10 outlines the opportunities and future challenges for xAI, leading to the conclusion in Section 11, where the key takeaways of the paper are summarized.

2. Overview of common terminology

2.1. Understanding interpretability versus explainability

As the reader may have noticed throughout this introduction, the terms explainability and interpretability are often used interchangeably. This leads to the definitions themselves being often generic and misleading (Rudin, 2018). The definitions of these terms are constantly subject to interpretation and debate, or even contradict each other, but this paper will utilize them in the same fashion as defined by Das and Rad (2020) and Mersha et al. (2024), wherein explainability and interpretability are regarded as interconnected yet separate ideas.

As outlined by Craig and Wisbey (2024) and Gilpin et al. (2018), interpretability refers to comprehending exactly how a model generates its output, while explainability involves conveying the behavior of a machine learning model in understandable, human terms. As a result of this definition, only certain models can be deemed interpretable, whereas explainability can be conceived for many non-interpretable models. This separates interpretability and explainability into two forms of xAI, when handling the aforementioned "white-box" models and the "black-box" models. "White-box" models, like decision-trees and rule-based frameworks, are inherently interpretable but they encounter the Interpretability-versus-Accuracy trade-off when compared to deep-learning models. This too showcases the inherent importance in realizing explainability, as it will enable the use of more accurate models.

2.2. Overview of post-hoc explainability techniques

Initially, a significant amount of the research in xAI techniques pertained to identifying patterns in how convolutional neural networks produce image classification predictions (Holzinger et al., 2020). This research extended to understanding the decision-making process of other types of complex neural network models, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs). Due to the ever-growing popularity of machine learning algorithms, emerging "black box" models tend to be highly varied in design. As a result, xAI techniques aim to be model agnostic, so as to be usable for any newly developed technology. As described by Arrieta et al. (2020), these techniques tend to rely on one of three general approaches: model simplification, feature relevance or visualization.

2.2.1. Interpretation using model simplification

Approaches using model simplification aim to approximate the working "black-box" model using a more readily explainable

surrogate model (Minh et al., 2022). Among these, one of the most used approaches today is known as Local Interpretable Model-Agnostic Explanations or LIME (Ribeiro et al., 2016). LIME is a local xAI technique which feeds randomized inputs based in a neighborhood around some initial input point x , to a complex model f_M . The outputs from f_M are used to estimate the local nature of the original machine learning (ML) model using an explainable linear surrogate function, f_s (Holzinger et al., 2020). Due to its use of the surrogate function, LIME is often referred to as a surrogate-based explanation technique (Samek et al., 2021). There are many additional model simplification techniques that use similar methods, such as GraphLIME which fits a non-linear surrogate function (Huang et al., 2020) and O. Bastani et al.'s method wherein a transparent function is developed to mimic the functionality of the complex function (Bastani et al., 2017). Besides these, there are rule setting techniques, wherein methods are developed to define specific rulesets that define the complex function's actions. U. Johansson et al. developed one such approach using a genetic programming algorithm, G-REX (Johansson et al., 2004). Each of these techniques are post-hoc explanations for a localized input set, as they rely on a finite input set to generate model output. The choice of the neighborhoods in which the input sets lie, will influence the design of the simplified model, and ultimately the quality of the xAI technique's interpretability.

2.2.2. Interpretation using feature relevance

Feature relevance approaches, or sensitivity analysis, aim to explain the functionality of a complex model by ranking its features' by their importance/relevancy for a specific output (Lin et al., 2020). One of the most used approaches in this area is known as SHapley Additive exPlanations, or SHAP (Lundberg & Lee, 2017). SHAP uses a method to calculate a feature importance score for each possible prediction, depending on the prediction's accuracy and consistency. Evidently, to use this approach, one must know the set of possible predictions and proper KPI metrics to determine accurate/consistent predictions. Nevertheless, as seen in Fig. 3, this method is also a commonly used approach. As with LIME, there are variations to the SHAP approach, such as KernelSHAP (Covert & Lee, 2021) and Asymmetric Shapley Values (ASV) (Frye et al., 2019). Other common feature relevance approaches include using Game Theory (Strumbelj & Kononenko, 2010) and local gradients (Baehrens et al., 2009) to test how changing features affects the output of the model. Compared to techniques using model simplification, feature relevance has comparable popularity, showing both approaches have significant traction today (Arrieta et al., 2020).

2.2.3. Interpretation using visualization

The final broad type of xAI techniques is visualization. These techniques allow humans to easily interpret model choices, and are normally related to decision tree-algorithms and computer vision techniques. Still, some visualization approaches aim to be model-agnostic, basing their visualizations on methods related to model simplification or feature relevance (Cortez & Embrechts, 2011). When used specifically to visualize complex decision-tree models, like XGBoost (Chen & Guestrin, 2016), there are typically ante-hoc (Holzinger et al., 2020) visualization methods built into the decision-tree model. When these diagrams become overly complex, these decision trees can be simplified by defining the decision tree based on input subsets (Domingos, 1998). When used specifically on CNNs for computer vision problems, attribution visualization techniques have been developed that highlight relevant input features. One such popular approach is the Gradient-weighted Class Activation Mapping (Grad-CAM) (Selvaraju et al., 2019). In many ways, these visualization approaches contribute similar results as the model simplification and feature relevance techniques but make the decision explanation easier to digest for visual problems. Visual explanations are less commonly used in the field, due to them being specifically valuable to tree-type models and models dealing with images, though they can be highly valuable when used in these situations (Alicioglu & Sun, 2022).

2.3. Overview of ante-hoc explainability techniques

As opposed to post-hoc techniques, ante-hoc techniques are employed as a part of the model design itself, and thus require access to the model's base structure. In other words, ante-hoc explainability refers to techniques where models are inherently interpretable by design. As mentioned, various "white-box" models follow these ante-hoc explainability methods, but certain deep-learning approaches have been realized in previous years, with varying levels of accuracy. For example, attention mechanisms can allow models to focus on the most relevant parts of the input data for models, improving their general explainability. Vaswani et al. (2017) proposed a powerful new method to focus on the most important parts of the input data. Researchers have found various methods to utilize this technique to dynamically weigh the importance of different input elements at each step of processing, helping deep-learning models like LSTMs (Bahdanau et al., 2014), BERT/GPT Transformers (Devlin et al., 2018) and Computer Vision based-models (Vision Transformers or ViTs) (Dosovitskiy et al., 2020) make more explainable decisions. However, it is crucial to note that attention weights do not necessarily indicate causal importance and may not always provide faithful explanations. As highlighted by Jain and Wallace (2019) in their extensive review, attention-based explanation models can exhibit inconsistencies and may fail to provide meaningful and accurate explanations for the model's ground predictions. Attention weights should therefore be validated using faithfulness checks or counterfactual tests to ensure their reliability as explanation mechanisms. Practitioners should be cautious when interpreting attention weights as direct explanations and consider them as one component of a broader explainability strategy rather than definitive causal indicators. Other ante-hoc methods such as Capsule Networks (CapsNets) (Sabour et al., 2017) or Neural Symbolic Networks (NSNs) (Besold et al., 2017), alter the foundational models (such as CNNs, in the case of CapsNets) in order to retain spatial information along the training and inference process, which makes them more interpretable in terms of recognizing and understanding the spatial properties of their input (i.e., images). Similar to the attention mechanisms, these methods that retain simplified information for the purpose of explainability are significantly more computationally intensive than their "black-box" counterparts, and often result in much lower accuracies due to their inner layer simplifications (Sabour et al., 2017). Generally speaking, many such ante-hoc methods prioritize interpretability, which can sometimes come at the cost of predictive performance. For instance, models designed to be interpretable may not reach the same accuracy levels as black-box models like deep CNNs or large transformers, and add a high level of complexity.

However, constant improvements to these approaches that make them more computationally-efficient and generally applicable, much like the step from SENN (Alvarez-Melis & Jaakkola, 2018) to QSENN (Norrenbrock et al., 2023), are continuing to occur.

Table 2 provides a comprehensive comparison of ante-hoc and post-hoc explainability approaches, highlighting their key characteristics, advantages, disadvantages, and appropriate use cases.

3. Detailed introduction to common explainable AI techniques

The previous sections have introduced the importance of explainability in artificial intelligence and highlighted some foundational concepts and methodologies. The following sections will now provide a detailed exploration of common Explainable AI (xAI) techniques. These techniques, ranging from model-specific methods to more generalized post-hoc approaches, are crucial for enhancing transparency, trust, and accountability in AI systems. By diving deeper into each technique, we aim to offer a clear understanding of how they work, their applications, and their strengths and limitations in different AI contexts. Given the rapidly developing landscape of xAI techniques, this paper focuses on presenting a diverse set of widely used methods. By providing a strong

Table 2

Comparison of ante-hoc and post-hoc explainability approaches, highlighting their definitions, examples, advantages, disadvantages, common pitfalls, and appropriate use cases.

Aspect	Ante-Hoc explainability	Post-Hoc explainability
Definition	Explainability built into the model architecture during design; models are inherently interpretable by design.	Explainability applied after model training; techniques analyze and explain already-trained models.
Examples	- Decision trees - Linear/Logistic regression - Rule-based systems - Attention mechanisms - Self-Explaining Neural Networks (SENNs) - Capsule Networks	- LIME - SHAP - Partial Dependence Plots (PDPs) - Grad-CAM - Counterfactual Explanations - Model Distillation
Advantages	- Inherent transparency - No additional computation for explanations - Guaranteed interpretability - Direct access to model internals - Can be more trustworthy	- Applicable to any trained model - Model-agnostic approaches available - No need to retrain models - Can explain complex black-box models - Flexible and adaptable
Disadvantages	- Often lower accuracy than black-box models - Limited to specific model architectures - Higher computational complexity during training - May require model redesign - Trade-off between interpretability and performance	- Explanations are approximations - Additional computational overhead - May not capture true model behavior - Potential for unfaithful explanations - Requires careful validation
Common Pitfalls	- Assuming attention weights indicate causal importance - Over-simplification leading to poor performance - Misinterpreting interpretability as accuracy guarantee - Computational inefficiency in complex models	- Assuming explanations are always faithful - Over-reliance on local explanations - Instability in explanation generation - Lack of validation of explanation quality
When Most Appropriate	- High-stakes applications requiring guaranteed transparency - Regulatory compliance needs - When model accuracy can be slightly sacrificed - When direct model inspection is required - Simple to moderate complexity problems	- Complex black-box models already in production - When model accuracy is paramount - Need to explain existing models without retraining - Large-scale or high-dimensional problems - When flexibility in explanation methods is needed

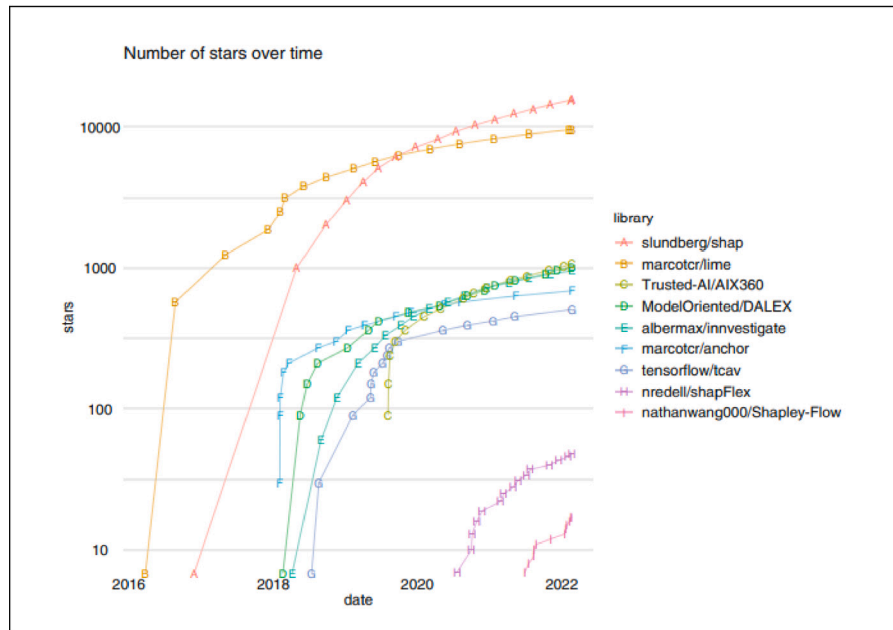


Fig. 3. Count of stars on GitHub for the most popular xAI repositories since 2016.

foundational understanding, readers will be better equipped to grasp future advancements, many of which may build upon the approaches discussed here. We will begin with several common techniques, the first two of which are clearly the most used over the last few years, according to Fig. 3.

Fig. 4 showcases the evolution of some of these prominent xAI techniques today, along with the evolution of significant AI modeling techniques.

3.1. LIME

LIME (Local Interpretable Model-agnostic Explanations) is a technique proposed by Ribeiro et al. (2016) to provide explainability for black-box deep learning models, making them more comprehensible for human experts. The primary objective is to foster trust and improve the decision-making process with machine learning models. LIME focuses on explaining individual predictions by generating a locally

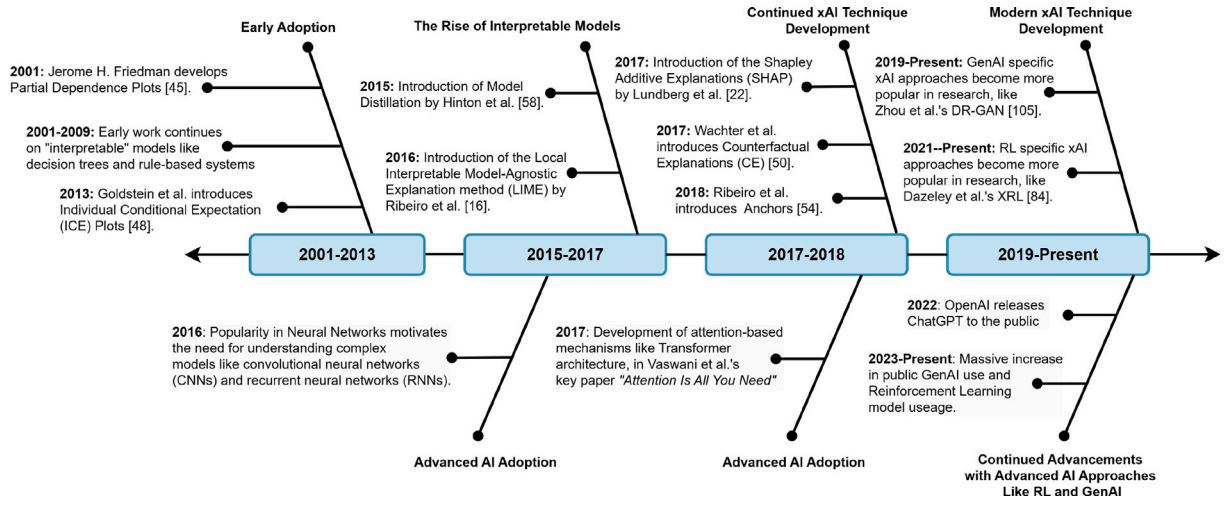


Fig. 4. Timeline showcasing the evolution of advanced AI modeling techniques and relevant corresponding xAI approaches.

interpretable approximation to the black-box model. It works by perturbing the input, collecting the predictions, and fitting a linear model on the perturbed dataset around a prediction under examination. The interpretable model then explains the specific prediction provided by the black-box model in terms of feature contribution, making it easier for humans to understand the prediction process. More explicitly, the LIME method involves the following steps (Ribeiro et al., 2016):

1. Data Perturbation: For an instance x_0 , LIME generates a set of perturbed samples $S = \{x_1, x_2, \dots, x_n\}$ by randomly modifying the features of x_0 . These perturbations are chosen based on the domain of the data (e.g., for text, word removal or synonym substitution might be used).

2. Local Surrogate Model Training: After obtaining predictions from the black-box model for each perturbed sample, a surrogate model g is trained. The goal of g is to be interpretable and to approximate the decision boundary of the black-box model f in the neighborhood of x_0 . The training of g is done by minimizing the weighted loss function:

$$\mathcal{L}(g) = \sum_{i=1}^n \pi(x_i, x_0) \cdot \text{Loss}(g(x_i), f(x_i)),$$

where:

- $\pi(x_i, x_0)$ is a proximity weight function that assigns higher weights to perturbed samples x_i that are closer to the instance x_0 .
- $\text{Loss}(g(x_i), f(x_i))$ is the loss function comparing the predictions of the surrogate model g with the black-box model f .

3. Interpretable Explanation: The surrogate model g provides a locally faithful explanation of the black-box model's behavior near the instance x_0 . For instance, if g is a linear model, the coefficients of g can be used to explain which features of x_0 are most important for the prediction.

Applying LIME to deep learning forecasting models can help in determining the most influential features for forecast predictions and understanding the model's decision-making process. For instance, LIME was employed to explain financial distress prediction models (Park et al., 2021) where it helped identify influential features for prediction, leading to actionable insights for users. Ultimately, LIME is a highly useful xAI technique due to its model agnostic nature, but it can have high instability due to its inability to provide global model explanations. In conclusion, LIME is a highly valuable technique in increasing the transparency of deep-learning forecasting models and provides valuable local insights for various kinds of models.

3.2. Shapley values

Another highly utilized, model agnostic xAI technique, Shapley Values, is inherently derived from cooperative game theory. It has been increasingly used to interpret and explain deep-learning forecasting models. This method, first introduced by Lloyd Shapley in 1953, gives a value to each feature in the model by considering their contribution to the model's output for a particular instance (Shapley, 1953). Then, some 59 years later, nuclear physicist-turned-economist Alvin Roth was awarded the Nobel Prize in Economic Sciences (2012) for expanding Shapley's work (Outreach, 2012). The technique has attracted significant interest due to its well-established theoretical foundation. In the context of deep-learning models, Shapley Values provide an effective and consistent way to quantify and rank feature importance for a specific prediction, rather than the standard global feature importance often derived from traditional statistical methods (Lundberg & Lee, 2017). The application of Shapley Values to deep-learning forecasting models has proven beneficial in various fields, such as energy consumption, finance, and healthcare. Ultimately, it enables the extraction of actionable insights from these typically complex models, helping to optimize resource utilization and processes, while ensuring fairness and equitable distribution of the benefits or risks tied to the predictions. Research specifically conducted by Lundberg et al. underscores the efficacy of Shapley Values in explaining complicated neural network-based forecasting models, such as LSTMs (Long Short-Term Memory) (Lundberg et al., 2019). They have demonstrated that by using Shapley Values, researchers and practitioners can better understand the essential features driving the model's forecasts, which may lead to improvements in model performance, decision-making, and trust in model outputs.

3.3. Partial Dependence Plots (PDP)

Partial Dependence Plots (PDPs) are a powerful and explainable technique for understanding feature importance from complex machine learning models. Developed by Jerome H. Friedman in 2001 (Friedman, 2001), PDPs aim to quantify the marginal effect of specific features on model predictions, capturing the relationship between an input feature and the output predictions while accounting for other input features' average effects. More simply, they help explainability by providing insights into how a particular feature or a combination of features influences the model's prediction. They overcome the limitations of feature importance based on model parameters by allowing non-linear interactions between features and enabling visualization of feature effects on model predictions in response to local variation (Kim et al., 2016).

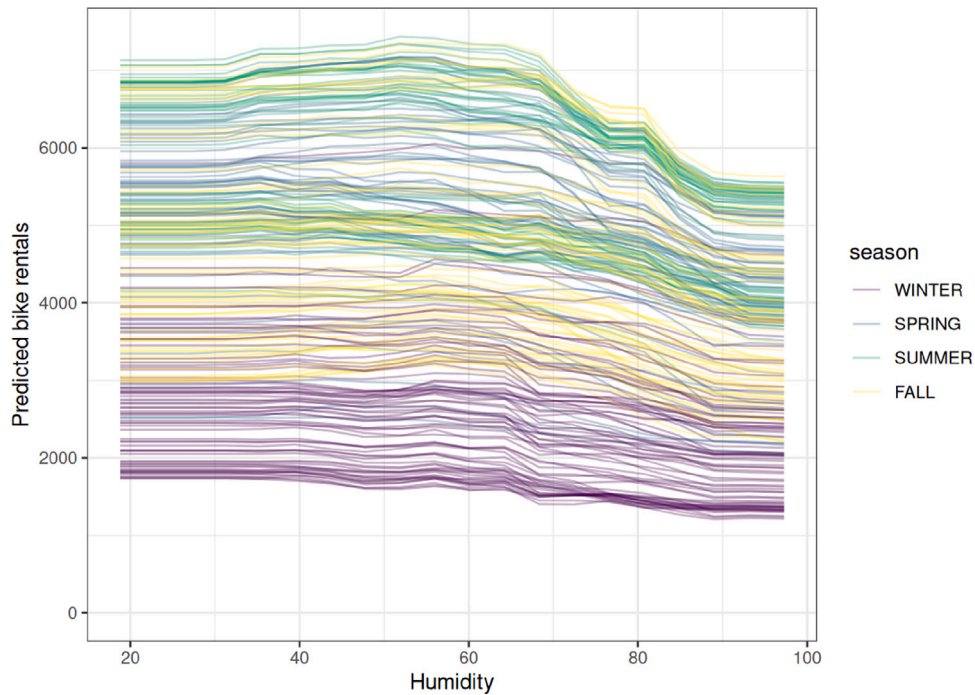


Fig. 5. ICE curves for a random forest predicting bike rentals, with humidity and season as model inputs.

In the context of deep-learning forecasting models, PDPs are especially valuable for time series data, given that these models involve multiple input features that might interact with each other in highly complex and non-linear ways. PDPs identify the most significant features driving model predictions and reveal potential biases or issues with the dataset that may affect the forecast. By visualizing and analyzing PDPs, practitioners can gain insights into how a specific feature impacts the forecasted outcome and make decisions to improve model performance, robustness, and generalization (Molnar, 2019). As an example of an implementation, Goldstein et al. (2015) applied PDPs for time series forecasting and demonstrated their effectiveness in explaining deep-learning models' predictions. However, they demonstrated that these classical PDPs can potentially obfuscate the complexity of the modeled relationship between the predicted response and the input features, thus highlighting a limitation of the PDP approach. Furthermore, Goldstein et al. demonstrated that Individual Conditional Expectation (ICE) Plots can alleviate some of these limitations and shed light on estimated models in ways that PDPs cannot (Goldstein et al., 2015). In summary, these Partial Dependence Plots provide a valuable xAI technique for understanding how specific features contribute to model predictions and enable better comprehension of deep-learning forecasting models' behaviors. They offer practitioners a powerful and explainable visual analysis tool for improving model performance and ensuring the robustness of machine learning models in time series forecasting.

3.4. Individual Conditional Expectation (ICE) plots

Similar to PDPs, Individual Conditional Expectation (ICE) Plots are an advanced explainable machine learning technique that enables the visualization and explanation of complex models such as deep-learning forecasting systems. This method, introduced by Goldstein et al. (2015), also provides insights into the effects of input features on the model's output by plotting the predictions for individual instances while varying a specific feature. The ICE plots produce intermediary partial dependence plots that help uncover how an input feature influences the outcome at different levels. The plot provides valuable information about the directionality and heterogeneity of the relationship between an input variable and the response (Molnar, 2019). Unlike PDPs, ICE

plots allow for visualizing how the effect of a feature changes for different instances, which can reveal interactions between features that might be missed by PDPs. In the context of deep learning forecasting models, ICE can be particularly beneficial in understanding the complex nonlinear relationships between input features and predictions, which tend to be typical in these models. When applied to time series prediction tasks, for example, ICE plots can reveal the sensitivity of the model output to changes in input features. This understanding allows domain experts to explain the predictive behavior of deep learning models better and fosters trust in their predictions (Montavon et al., 2017). Moreover, ICE can help identify potential biases in the model and, subsequently, improve model explainability, robustness, and performance by enabling users to adjust the identified feature weights accordingly. Fig. 5 shows an example of this plot, where the inputs of Humidity and Season are graphed to show their impact on anticipated bike rentals. Despite the growing interest in applying ICE to deep learning forecasting models, it is important to understand one of its primary limitations: the computational cost associated with creating ICE plots, especially for high-dimensional data and large neural networks. Furthermore, interpreting ICE plots might be less intuitive for users with limited technical expertise, and for visualizations in higher dimensions. When there are many instances, the plot can become cluttered, making it hard to distinguish the individual curves. Aggregation methods like PDPs or smoothing techniques are often used to address this, and as with all of these techniques, future research in the field of xAI will continue to focus on overcoming these limitations to democratize the application and interpretation of ICE in deep learning models.

3.5. Counterfactual explanations

Next, there is Counterfactual Explanations (CE), which is a popular xAI technique that provides causal explanations for the predictions made by machine learning models. CE is a local explainability technique that relies on generating counterfactual instances by perturbing the input features while satisfying specific constraints, assuming that the model's prediction would have been different if those instances were observed (Wachter, Mittelstadt, & Floridi, 2017). According to Wachter, Mittelstadt, and Floridi (2017), counterfactual explanations help users understand the model's decision-making process

by expressing why a specific outcome occurred and explaining what minimal changes to the input features would have led to a different outcome. In deep-learning forecasting models, CE is particularly valuable due to the inherent complexity and opacity of these models. Verma et al. (2020) introduced a counterfactual generation approach using an optimization framework that applies sparsity-inducing regularization and constraints. This method can be adapted for forecasting models, enabling interpretable adjustments in real-world applications. In practice, Counterfactual Explanations (CE) have been applied to provide actionable insights to loan applicants, explaining why a user's application was rejected and what changes would have led to approval. Poyiadzi et al. (2019) introduced Counterfactual Paths, which was a slight tweak to traditional CE, to generate realistic and feasible changes an applicant could make over time to improve their chances of loan approval. Instead of suggesting abrupt, unrealistic modifications (e.g., "increase income by 50% instantly"), their approach considers gradual and actionable steps, such as building a credit history or reducing debt. This method has practical implications in finance, ensuring that model-driven decisions are not only transparent but also provide meaningful guidance to users seeking to improve their outcomes.

Ultimately, integrating CE into deep-learning forecasting models improves explainability and trustworthiness by revealing the key features that drive predictions. It also aids in model validation, as counterfactual explanations can be compared against domain knowledge, facilitating refinement and fine-tuning (Gunning et al., 2021). Additionally, CE supports informed decision-making by allowing stakeholders to explore alternative scenarios and potential interventions.

3.6. Anchors

Anchors, introduced by Ribeiro et al. (2018), is an advanced method for explaining predictions of "black box" models by presenting the features of the instance that are most fundamental for prediction. The method uses a reinforcement learning process wherein decision rules are acquired based on the learned importance of the different features. Anchors generate simple and interpretable explanations, promoting confidence, fairness, and preserving users' trust in the model (Gunning & Aha, 2019). As with all of the techniques explained so far, this too is designed to be model-agnostic (Ribeiro et al., 2018), meaning it can be used to explain predictions produced by models of varying explainability, from decision trees to deep learning models. Additionally, it measures local explainability, meaning it is based on perturbations of various inputs to explain how the model behaves within a window of that input.

This approach is built on creating "anchors", which are if-then decision rules that reliably explain the model's predictions (Ribeiro et al., 2018). This technique works by perturbing the input data to find feature conditions that hold with high confidence across similar instances, ensuring that the explanation is both accurate and understandable. Unlike other methods, such as LIME, which provide approximate explanations, Anchors guarantee that the rule applies to a large proportion of similar cases, making it particularly useful for explaining complex models in a transparent and explainable way. Ultimately, its intent is to generate these "anchor" if-then statements to describe model actions. As a result, for highly complex models, it may be difficult to fully interpret their behavior with this approach.

An important advantage of Anchors is its focus on the prediction's local fidelity, which is the explanation's accuracy in a small region around the instance being explained. This approach enhances the method's transparency and helps users understand the factors that lead to a specific prediction (Ribeiro et al., 2016). Experiments conducted by Ribeiro et al. (2018) demonstrated the potential of Anchors for providing accurate and interpretable explanations on both text and tabular datasets, outclassing the previously established explanation technique LIME (Samek, 2019). Despite these advantages, there are still potential limitations to using Anchors. Recent studies have pointed

out its vulnerability to adversarial examples (Rudin & Radin, 2019), which indicate that explanations generated by Anchors may depend on the model's quality and its robustness to adversarial attacks. Further research is needed to test and improve Anchors on diverse data types and models to ensure its viability and reliability as a comprehensively useful approach to machine learning model explainability.

3.7. Model distillation

The final, widely applicable xAI technique that we will discuss in this section is that of Model Distillation. This is an effective technique used to transfer knowledge from a larger teacher model to a smaller student model while preserving the performance of the original model. The idea behind model distillation is to leverage the generalization capabilities of the teacher model, which is usually a network with greater capacity and has been trained on a large-scale, complex dataset. The student model, on the other hand, is a smaller, computationally efficient network designed for specific tasks or deployment scenarios where computational resources or memory is limited (Hinton et al., 2015). The key to model distillation lies in the training process of the student model. Instead of mimicking hard label predictions, the student model is trained to mimic the probability distribution of the teacher's output in a process called soft target learning (Zagoruyko & Komodakis, 2016). Using a loss function like the Kullback-Leibler (KL) divergence, the student model can approximate the outputs of the teacher model more efficiently, thereby preserving the inherent knowledge of the larger model. The advantages of model distillation include the student model's reduced size and computational costs while offering competitive performance to that of the teacher model. This efficiency makes it particularly suitable for applications such as edge devices and low-power embedded systems. Moreover, model distillation can be employed in various deep learning architectures such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and transformers (Romero et al., 2014). Model Distillation has gained considerable attention and has been widely explored in many research areas, including computer vision, natural language processing, and reinforcement learning. Challenges and future directions involve developing better loss functions, more effective distillation strategies, and finding an optimal teacher-student model combination to further improve the performance and efficiency of distilled models (Polino et al., 2018) (see Fig. 6).

4. Explainable AI techniques related to supervised learning techniques

Taking a step away from general approaches to Explainable AI (xAI), this next section will focus on techniques that can be applied to specific model types. Firstly, we can focus on supervised machine learning techniques.

When selecting an xAI approach for a supervised learning model, several factors must be considered to ensure that the chosen method effectively explains the model's behavior. First, the complexity of the model is crucial—simpler models such as decision trees or linear regression can often be adequately explained using methods like Partial Dependence Plots (PDPs) (Friedman, 2001), Counterfactual Explanations (Wachter, Mittelstadt, & Russell, 2017), or LIME (Ribeiro et al., 2016), which provide intuitive visualizations of feature importance. In contrast, more complex techniques such as Layer-wise Relevance Propagation (LRP) (Bach et al., 2015) or Grad-CAM (Selvaraju et al., 2019) are designed to handle the intricate relationships and non-linearity inherent in more complex models. Additionally, the level of explainability required is important, and the desired output must be carefully considered, as some applications may demand local explanations for individual predictions (e.g., LIME or SHAP) while others might benefit from global explainability, which explains overall model behavior (e.g., PDPs or SHAP values) (Yang et al., 2023). Data characteristics,

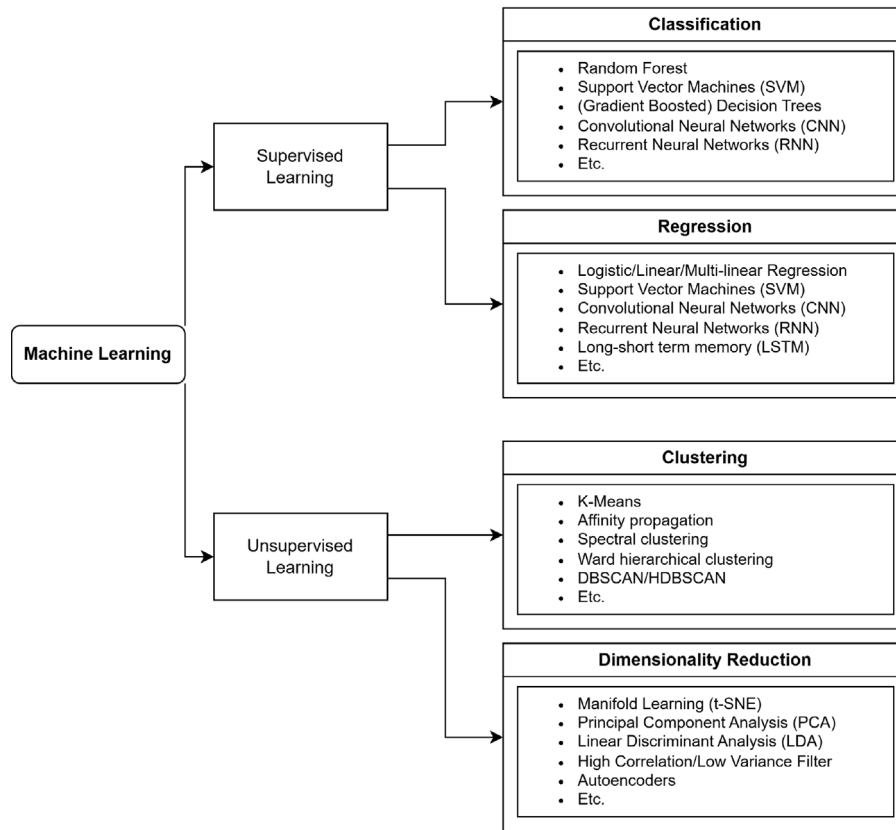


Fig. 6. Types of machine learning models, distinguished by Supervised or Unsupervised Learning approaches.

such as the presence of outliers or noisy features, also play a role in determining which method will provide the clearest insights. Finally, the goal of the explanation must be considered—whether the purpose is to ensure fairness, improve trust, or simply debug the model. Tailoring the xAI approach to the specific needs of the model and its application will lead to more effective and actionable insights.

4.1. Hybrid xAI approaches for supervised learning

Recent advances in explainable AI have led to the development of hybrid approaches that combine multiple explanation methods to provide more comprehensive and robust explanations for supervised learning models (Álvarez et al., 2024). These hybrid methods address the limitations of individual techniques by integrating complementary explanation strategies, often combining local and global perspectives to offer a more complete understanding of model behavior.

One prominent category of hybrid approaches combines local and global explanations to provide both instance-specific insights and overall model understanding. For example, methods that integrate LIME with SHAP values can provide local explanations for individual predictions while also offering global feature importance rankings. Similarly, approaches that combine counterfactual explanations with partial dependence plots enable practitioners to understand both what would need to change for a different outcome (local) and how features generally affect predictions across the dataset (global) (Yenikar et al., 2025). These hybrid methods are particularly valuable in applications where stakeholders need both detailed explanations for specific decisions and a broader understanding of model behavior, such as in manufacturing (Yenikar et al., 2025), healthcare diagnostics or financial risk assessment.

Another category of hybrid approaches integrates multiple explanation techniques to enhance explanation quality and reliability. For

instance, ensemble explanation methods combine outputs from multiple xAI techniques (e.g., LIME, SHAP, and feature importance) to generate consensus explanations that are more stable and trustworthy than individual methods alone (Álvarez-Melis & Jaakkola, 2017). These approaches often use voting mechanisms or weighted aggregation to synthesize explanations from different techniques, reducing the impact of instabilities inherent in individual methods. Additionally, some hybrid approaches combine model-agnostic techniques (like LIME) with model-specific methods (like LRP for neural networks) to leverage the strengths of both approaches, providing explanations that are both flexible and faithful to the model's internal structure.

Scalable hybrid approaches have also emerged to address the computational challenges of applying xAI to large-scale supervised learning applications. These methods often employ sampling strategies, approximation techniques, or distributed computing frameworks to make hybrid explanations computationally feasible for large datasets and complex models. For example, scalable variants of SHAP use approximation algorithms or feature grouping to reduce computational complexity while maintaining explanation quality (Covert & Lee, 2021). Similarly, distributed implementations of hybrid explanation methods enable the application of xAI techniques to big data scenarios where traditional methods would be computationally prohibitive.

4.2. Scalability considerations for xAI in supervised learning

The scalability of xAI techniques is a critical consideration when applying explainability methods to supervised learning models in real-world settings. Computational requirements, applicability to different problem sizes, and trade-offs between explanation quality and efficiency must be carefully evaluated to ensure practical deployment.

Computational requirements vary significantly across different xAI techniques. Local explanation methods like LIME typically require

multiple model evaluations (often hundreds or thousands) for each explanation, making them computationally expensive for large-scale applications. Global methods like SHAP can be even more demanding, with computational complexity that grows exponentially with the number of features in the worst case. In contrast, simpler methods like feature importance scores or partial dependence plots are more computationally efficient but may provide less detailed explanations. Hybrid approaches that combine multiple methods can compound these computational costs, though recent advances in approximation techniques and efficient implementations have made hybrid methods more practical for large-scale applications.

The applicability of xAI techniques to different problem sizes presents another important scalability consideration. For small to medium-sized datasets (thousands to tens of thousands of samples), most xAI techniques are computationally feasible and can provide comprehensive explanations. However, as dataset size increases to hundreds of thousands or millions of samples, computational costs become prohibitive for many methods. In such cases, scalable approaches such as sampling-based methods, approximate algorithms, or distributed computing frameworks become necessary. Similarly, the number of features significantly impacts scalability: high-dimensional problems (hundreds or thousands of features) require specialized techniques or dimensionality reduction strategies to make explanations computationally tractable.

Trade-offs between explanation quality and computational efficiency are inherent in xAI applications. More detailed and faithful explanations typically require greater computational resources, while faster methods may sacrifice some explanation quality or completeness. For instance, exact SHAP values provide theoretically optimal explanations but are computationally expensive, while approximate methods like KernelSHAP or TreeSHAP offer faster computation at the cost of some approximation error. Similarly, local explanation methods can provide detailed instance-level insights but may not scale well to explaining large numbers of predictions, whereas global methods offer efficient overall model understanding but may miss important local variations.

The benefits of hybrid methods must be weighed against their increased computational requirements. While hybrid approaches can provide more comprehensive and robust explanations by combining multiple techniques, they often require significantly more computation than individual methods. However, the improved explanation quality and reliability may justify the additional computational cost in high-stakes applications where explanation accuracy is paramount. Recent research has focused on developing efficient hybrid methods that maintain explanation quality while reducing computational overhead through techniques such as intelligent sampling, caching, and parallel processing (Yang et al., 2023). As the field continues to evolve, scalable hybrid xAI approaches represent an active area of research, with ongoing efforts to balance explanation comprehensiveness with computational feasibility for large-scale supervised learning applications.

5. Explainable AI techniques related to unsupervised learning techniques

Similarly to the supervised approaches, Explainable AI (xAI) techniques have become increasingly important to make unsupervised machine learning methods more transparent, explainable, and trustworthy. In unsupervised learning, algorithms such as clustering, dimensionality reduction, and anomaly detection are used to discover hidden patterns in data without labeled examples. One prominent xAI technique in unsupervised learning is the use of visualization methods to understand learned data representations, such as t-Distributed Stochastic Neighbor Embedding (t-SNE) (van der Maaten & Hinton, 2008). t-SNE is widely used to visualize high-dimensional data by reducing them to two or three dimensions for easier interpretation. Another explainable AI approach is the development of meaningful feature

extraction and selection. Techniques such as autoencoders and sparse coding can be applied to learn compact and informative features in an unsupervised manner, which can then be analyzed and interpreted by domain experts (Ayinde & Zurada, 2018; Gao et al., 2024). Additionally, using decision trees and rule-based techniques has also shown to provide explainable insights in unsupervised learning, such as Bayesian rule lists (Letham et al., 2015). Overall, integrating explainable AI techniques into unsupervised learning methods allows for better understanding, trust, and validation of the results.

5.1. t-Distributed Stochastic Neighbor Embeddings (t-SNE)

t-Distributed Stochastic Neighbor Embedding (t-SNE) is a dimensionality reduction technique that has gained significant attention for visualizing high-dimensional data in a low-dimensional space, while preserving the local structure of the original dataset. The main idea behind t-SNE is to transform the data points in the high-dimensional space into low-dimensional embeddings by minimizing the discrepancy between two probability distributions. These probability distributions are derived from pairwise similarities in the high and low-dimensional spaces, respectively (van der Maaten & Hinton, 2008). Specifically, t-SNE uses Gaussian distribution to measure similarities in the high-dimensional space and t-distribution in the low-dimensional space, which further aids in alleviating the “crowding problem” commonly encountered with linear methods (van der Maaten & Hinton, 2008). Moreover, t-SNE employs gradient descent minimization of the Kullback–Leibler divergence in the embedded space (van der Maaten, 2014). Due to these unique characteristics, t-SNE has been widely employed in various unsupervised learning applications such as image recognition, natural language processing, and gene expression analysis (Wattenberg et al., 2016). Researchers have also expanded upon t-SNE with various adaptive and scalable extensions to handle large-scale datasets (van der Maaten, 2009).

It is crucial to understand that t-SNE should be framed as an *exploratory visualization tool* rather than a direct explanation method for unsupervised learning models. While t-SNE can provide valuable visual insights into high-dimensional data structures, its outputs can be misleading if not interpreted with appropriate caution and validation. The technique is highly sensitive to hyperparameters, particularly the perplexity parameter, which controls the balance between preserving local and global structure. Different perplexity values can produce dramatically different visualizations of the same dataset, potentially leading to incorrect interpretations about cluster structure or data relationships (van der Maaten, 2009). Additionally, t-SNE is sensitive to random initialization (seed sensitivity), meaning that different runs with the same parameters may produce different embeddings, further complicating interpretation.

Practitioners should perform robustness checks when using t-SNE, including: (1) varying the perplexity parameter across a range of values (typically 5 to 50) to assess stability of the visualization, (2) testing multiple random seeds to evaluate consistency of results, (3) comparing t-SNE embeddings with other dimensionality reduction techniques (e.g., PCA, UMAP) to validate findings, and (4) using quantitative cluster validation metrics (e.g., silhouette score, adjusted rand index) rather than relying solely on visual inspection. Importantly, t-SNE visualizations do not prove cluster validity—the presence of apparent clusters in a t-SNE plot does not necessarily indicate meaningful structure in the original high-dimensional space. The technique can create artificial clusters or merge distinct groups, making it essential to validate any apparent structure using independent clustering algorithms and validation metrics.

In the context of explainable AI (xAI), t-SNE can play a supporting role by providing visual interpretations of high-dimensional data representations, making it easier for domain experts to explore and understand learned data structures. However, it should be used as one component of a broader explainability strategy, complemented

by other unsupervised xAI techniques and validated through quantitative methods. When used appropriately with proper robustness checks and validation, t-SNE can help bridge the gap between complex machine learning models and human understanding, but practitioners must remain aware of its limitations and potential for misleading interpretations.

5.2. Explainable clustering methods

Beyond visualization techniques, several concrete unsupervised xAI methods provide interpretable explanations for clustering and pattern discovery. One prominent approach involves explainable clustering methods that generate human-readable descriptions of clusters, enabling users to understand why data points are grouped together and what characteristics define each cluster. These methods go beyond simply identifying clusters to provide interpretable rules, feature importance rankings, or natural language descriptions that explain cluster membership and structure.

For instance, explainable k-means variants incorporate feature importance analysis to identify which features are most influential in determining cluster assignments, providing transparency into the clustering process (Kim et al., 2016). Similarly, methods that generate interpretable cluster descriptions using decision trees or rule-based systems allow practitioners to understand cluster boundaries and characteristics in human-readable formats. These approaches are particularly valuable in applications where understanding the rationale behind clustering decisions is crucial, such as customer segmentation in marketing, patient grouping in healthcare, or anomaly detection in cybersecurity. By providing explicit explanations of why data points belong to specific clusters, these methods enhance trust and enable domain experts to validate and refine clustering results.

5.3. Bayesian rule lists

Bayesian rule lists represent another concrete unsupervised xAI exemplar that provides interpretable explanations beyond visualization. Bayesian rule lists are human-readable explanations that combine decision trees and ordered lists of if-then rules using probabilistic modeling methods focused on capturing interpretable patterns in the dataset (Yang et al., 2016). The technique works based on Bayesian statistics, where each rule is assigned a certain probability, and a Bayesian score is calculated for every rule list (i.e., combinations of rules) using prior knowledge and data likelihood (Kuhn & Johnson, 2013). To explore the space of possible rule lists, tools such as Monte Carlo Tree Search (MCTS) and Markov Chain Monte Carlo (MCMC) sampling methods have been widely used (Bengio, 2009). The resulting Bayesian rule list is not only easy to understand but also provides a robust probabilistic prediction model for various clustering and classification tasks (Lakkaraju et al., 2016; Zhang et al., 2016). This makes Bayesian rule lists a powerful and interpretable unsupervised machine learning approach for identifying latent structures and correlations within complex data sets.

In the context of explainable AI (xAI), Bayesian rule lists have gained popularity because of their transparency and interpretability. Their rule-based format allows both technical and non-technical users to understand how predictions are made, making them an excellent tool for explaining unsupervised machine learning outcomes. Moreover, the probabilistic nature of Bayesian rule lists allows for the quantification of uncertainty, which enhances their value in safety-critical applications where understanding model confidence is crucial. By integrating Bayesian rule lists with other xAI techniques, we can improve the transparency of unsupervised learning models, enabling users to understand both the patterns identified by the model and the uncertainty surrounding those patterns.

6. Explainable AI techniques related to reinforcement learning techniques

Reinforcement learning (RL) methods are increasingly popular due to their heavy adaptability and power when handling highly complex problems. Unfortunately, this results in RL methods having high complexity, and understanding their functionality is quite difficult. The primary objective of xAI in the context of RL is to generate understandable and explainable explanations of the policies learned by agents and provide insights into their underlying reasoning process (Arrieta et al., 2020).

A popular approach for achieving explainability in RL is the use of attention-based mechanisms, which allow identifying the key features contributing to the agent's decisions (Vaswani et al., 2017). This is valuable for both developers and domain experts, as it provides a clearer understanding of how agents prioritize different aspects of their environment when making decisions. Another prominent technique involves employing saliency maps, which visually highlight the regions in the input space that are crucial for decision-making (Greydanus et al., 2017). This visualization allows stakeholders to track and validate the agent's decision-making process in a more intuitive manner. Recent research has also proposed the use of local linear models for providing more transparent explanations of the agent's behavior by approximating the learned policy with simpler, human-readable models (Miller, 2017). Such models help bridge the gap between complex RL techniques and practical application, offering clearer insights into how specific decisions were made.

Moreover, inverse reinforcement learning (IRL) can be employed to extract interpretable reward functions from expert demonstrations, allowing for the analysis and understanding of the expert's decision-making process (Abbeel & Ng, 2011). IRL adds significant value in applications where human expertise must be encoded into agent behavior, improving model accountability. In the context of deep RL, research has also focused on the application of layer-wise relevance propagation methods to attribute the agent's decisions to specific input features (Bach et al., 2015). This technique is essential for understanding how deep neural networks in RL contribute to complex decision-making, ensuring that the decisions made by the agent are justifiable and aligned with the task's goals.

These various xAI techniques enable reinforcement learning models to generate reliable and explainable policies, contributing to safer and more trustworthy AI systems in practical applications. By improving the explainability of RL agents, xAI fosters greater accountability and transparency, which are critical for real-world adoption in safety-critical domains, such as healthcare, finance, and autonomous systems.

6.1. Attention-based mechanisms

Attention-based mechanisms have gained prominence in the field of explainable Artificial Intelligence (AI) by enhancing the explainability of deep reinforcement learning (RL) techniques. The primary goal of attention mechanisms is to provide the AI agent with the ability to recognize and focus on relevant aspects of the environment when making a decision, thereby improving the explainability of its learned policy. In attention-based reinforcement learning, the agent's decision-making process is guided by a hierarchical structure where the attention mechanism acts as a weighting function, enabling the agent to concentrate on vital components while filtering out less critical information (Manchin et al., 2019). One key advantage of attention-based mechanisms is their capability to handle large state and action spaces effectively. By promoting the interpretation of temporal relationships between actions and states, these mechanisms enhance the agent's ability to extract valuable information from past experiences, thereby improving its overall performance in complex environments (Bahdanau et al., 2014). In addition, attention mechanisms can significantly augment the transferability and generalization of learned policies to novel

tasks and settings, providing a strong basis for incorporating human-like attention abilities in AI systems. Notably, attention-based models have been successfully applied to various RL problem domains, such as natural language processing, computer vision, and robot control (Mnih et al., 2014). Overall, attention-based mechanisms serve as a vital component for advancing explainable AI by offering a more transparent and human-understandable solution to reinforcement learning techniques.

6.2. Saliency mapping

This next approach, saliency mapping, has gained substantial attention in the context of reinforcement learning (RL) algorithms. As a visualization technique, saliency mapping identifies and quantifies the importance of specific input features to the model's decisions, providing users with insights into how the agent arrives at its predictions or actions. By perturbing the input features and observing the corresponding changes in the output, saliency maps generate a heatmap that highlights which features are most influential in the agent's decision process (Greydanus et al., 2017). This technique becomes particularly valuable in RL, where the complexity of decision-making in environments can be opaque and difficult to interpret.

In the context of RL, saliency maps offer researchers the opportunity to probe the internal workings of deep RL agents. They can identify the most critical factors influencing the agent's behavior, uncover potential blind spots, and diagnose causes for unexpected or suboptimal decisions. For example, in RL applications like AlphaGo Zero (Silver, Hubert, et al., 2017), saliency mapping techniques have been employed to gain a deeper understanding of the agent's policy and value functions, shedding light on its strategic and tactical decision-making in complex games (Silver, Schrittwieser, et al., 2017).

The importance of saliency mapping for explainable AI (xAI) in RL lies in its ability to improve model transparency and trustworthiness. By providing explainable visualizations of how certain input features impact decisions, saliency maps not only help to debug RL models but also enable practitioners to verify the rationality of the learned policies. This is especially vital in safety-critical domains, where ensuring model reliability and identifying unforeseen behaviors can significantly impact the outcomes of RL applications. Ultimately, saliency mapping is a key tool for making RL models more transparent, explainable, and dependable in real-world scenarios.

6.3. eXplainable Reinforcement Learning (XRL)

Another area of research tying xAI with RL is research from Dazeley et al. showcasing a Broad-xAI approach that aims to provide general explainability for RL models (Dazeley et al., 2021). Their paper explores the development of Explainable Reinforcement Learning (XRL) and its potential for building a broader concept of Explainable AI (Broad-xAI). The authors introduce the Causal XRL Framework (CXF), designed to integrate reinforcement learning (RL) with external ontologies and communication systems, enabling RL agents to justify decisions to users. XRL is outlined as a distinct subfield within Explainable AI, offering a conceptual framework for RL-based explanations. The primary benefit of XRL is its ability to provide transparent, human-readable justifications for RL decision-making, which increases trust and accountability, particularly in high-stakes domains like healthcare and finance. By incorporating causal reasoning, XRL can offer valuable insights into the cause-and-effect relationships within the model, helping users understand why a certain action leads to a specific outcome. This integration of external ontologies further enhances the flexibility and applicability of XRL across a wide range of domains. However, the approach is not without its challenges. One of the main drawbacks is the complexity involved in modeling causal relationships, which can make the system harder to implement, particularly in environments with numerous variables or dynamic elements. Additionally, the computational overhead required to generate causal explanations may limit

its use in real-time applications. Finally, it is an Ante-Hoc process, meaning it must be integrated into the core model's infrastructure, and cannot be applied in a Post-Hoc manner. XRL's dependency on external knowledge and ontologies also means that it may struggle with generalization, particularly in rapidly evolving fields where domain-specific information is constantly changing. Despite these limitations, XRL offers a promising pathway toward more transparent and interpretable AI systems, provided its complexities can be addressed.

6.4. Local-linear models

The final mentioned significant approach in reinforcement learning (RL), aimed at improving explainability, is the use of "local-linear models". This method approximates the true function of the model with locally linear surrogate models, providing a more transparent understanding of the decision-making process without sacrificing performance. The key idea is that, within specific regions of the input space, the function is assumed to be linear, enabling easier explanation of how input features influence predictions (Ribeiro et al., 2016).

For instance, Wulfmeier et al. (2015) applied local-linear models to inverse reinforcement learning (IRL) in the context of maximum entropy deep IRL, demonstrating their ability to learn more detailed feature functions and capture relevant environmental dynamics. This led to a more robust understanding of complex tasks, improving convergence and stability during the learning process compared to simpler, abstract feature functions.

Overall, local-linear models serve as a crucial tool for enhancing interpretability in RL. They allow researchers and practitioners to gain deeper insights into how features interact with predictions in different regions of the input space, thus contributing to more transparent and trustworthy AI systems. By improving interpretability, this technique fosters greater confidence in RL models, which is especially important in high-stakes applications such as robotics and autonomous systems.

7. Explainable AI techniques related to computer vision image detection models

Now, let us take a step back into Supervised Learning approaches, and focus in on Computer Vision models. The reason why these models require special attention, is due to their input/output data formatting, and how the explanation of prominent features can be best represented. Typically, explanation methods for image detection can be categorized into three major types: visual explanations, textual explanations and example-based explanations (van der Velden et al., 2022). Among these, visual explanations comprise the most used techniques, as seen in Fig. 3, although textual and example-based explanations can still be highly useful in certain cases.

7.1. Visual explanations

Also referred to as saliency mapping, visual explanations aim to highlight the most important parts of an image used in the model's decision-making process. Most visual explanation techniques tend to use backpropagation to identify heavily focused parts of the image, although there are also techniques that utilize perturbation-based approaches, such as LIME, or multiple instance learning-based approaches. These approaches will be further discussed in the upcoming sections.

7.1.1. Guided backpropagation

One of the first saliency mapping techniques used for identifying pixels in images that had the highest impact on the model's output, was

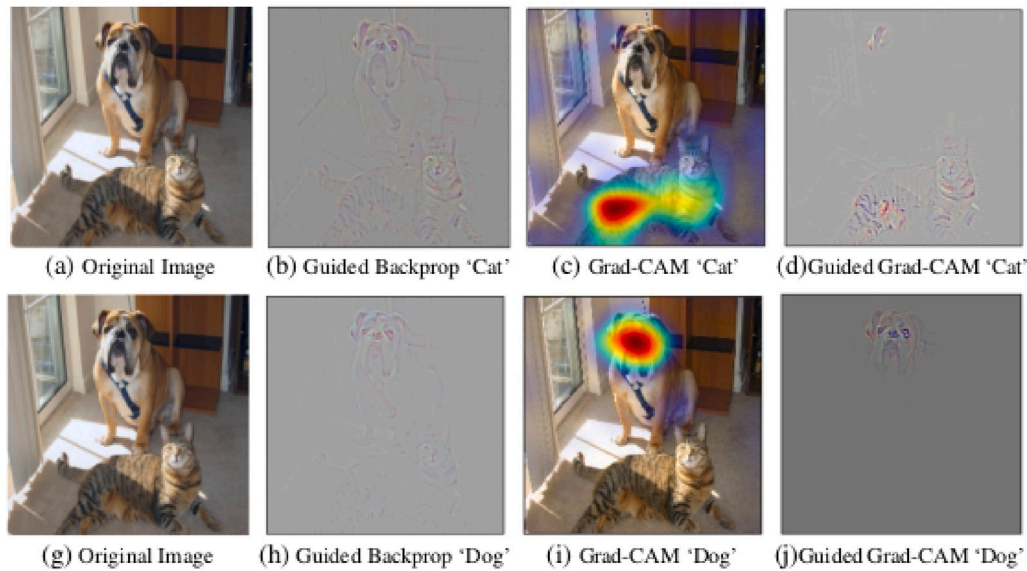


Fig. 7. Grad-CAM heat map results showing original images, guided-backpropagation results, Grad-CAM results, and guided Grad-CAM results for image object detection models identifying a cat and a dog.

by using guided backpropagation techniques. These techniques provide local, post-hoc explanation for convolutional neural networks (CNNs), which are a highly used modeling technique for object detection in images (Simonyan et al., 2013). Specifically, guided backpropagation is a technique that backpropagates through the Relu activation function in a CNN model, and identifies the positive gradient with respect to images (Springenberg et al., 2014). Through this process, one can visualize the image features that activate the neurons of the CNN model. The output of this approach, as with most visual explanation techniques, is images that reflect the input images in their dimension and background, but highlight the areas of the image that maximized the activation of the feature maps. As shown in Fig. 7, the gray areas of the output images represent areas of the image that did not contribute to the model's detection, whereas colored areas of the image did. Each of the images for the respective models represents the model's activation mapping at the different layers of the model. As a result, the final images give a better insight into how the model will behave once fully trained, but this approach also gives insight into exactly how these models are iteratively training. A major advantage of this approach is its ability to interpret activity in convolutional and fully connected layers of the model. Conversely, this approach does have a downside in that it is restricted to work only on CNN type models, and thus is not model-agnostic.

7.1.2. Grad-CAM

Another common visual xAI approach, that was previously mentioned in this paper, is known as class-activation mapping (CAM). This approach involves redefining the training structure of a CNN network, to add global pooling layers on the final convolutional feature mappings (Zhou et al., 2015). Similarly to the guided backpropagation technique, the output of this approach provides local explanations, though it is not model-agnostic. Following the creation of CAM explanation approaches, Selvaraju et al. introduced Gradient-weighted Class Activation Mapping (Grad-CAM) (Selvaraju et al., 2019), which generalized the use of the CAM approach. Grad-CAM allows the CAM approach to be used across various CNN networks, without introducing the complication of adding global pooling layers to the model itself. This allows the approach to become available as a post-hoc approach, which is extremely important when working with pre-trained models. In their same paper, the authors of Grad-CAM also introduced a method of Grad-CAM entitled guided Grad-CAM, which incorporates the results

from guided backpropagation and Grad-CAM. These approaches are heavily used in spaces such as medical imaging, where diagnoses for ailments or anomalies are of the utmost importance (van der Velden et al., 2022). In similar fashion to the guided-backpropagation approach, the output of any CAM, approach showcases an image with highlights overlayed on top of the original image, showcasing relevant features to the model's interpretation. Alternatively to the guided-backpropagation output, Grad-CAM tends to overlay a heat-map over a region of the image that it deems important to the model, rather than a pixel level explanation of important regions. As a result, the output from Grad-CAM often does not point to a specific feature in the image, but a region of the image that it deems important. An example these compared outputs can be shown in Fig. 8, which was taken from Selvaraju et al.'s paper. Luckily, as shown in Selvaraju et al.'s paper, weakly supervised segmentation approaches, such as one produced by Kolesnikov and Lampert (2016), can be used on top of Grad-CAM, to segment individual important objects and features in the image based on the results of the Grad-CAM results. An example of this segmentation can be seen in Malhi et al.'s work (Malhi et al., 2019).

7.1.3. Deep SHAP

The final back-propagation approach that will be discussed in this section is called Deep SHAP. Expanding on the previously explained Shapley Additive exPlanations (SHAP) technique, Deep SHAP combines them with a deep learning approach called DeepLIFT (Deep Learning Important Features) (Shrikumar et al., 2017). DeepLIFT is a method that decomposes the output prediction of neural networks by backpropagating the contributions of all neurons in the neural networks to every feature of the input. Using this output, Lundberg et al. devised a method to approximate Shapley values for these convolutional neural network models. Kernel SHAP approaches can then be used to interpret important features in the original image and, similar to the Grad-CAM output, the regions in the original image of high importance can be highlighted. This approach is also a local explanation, able to explain results for single inputs, but is again not model agnostic, as it is designed to work with CNN type models. This approach is different from the previous two defined back-propagation approaches as it ensures an independent analysis of each individual pixel (or voxel in the three-dimensional sense), and thus a non-biased assessment of attribution (van der Velden et al., 2020). Unfortunately, as previously discussed, Shapley values can be highly costly to compute. As Deep SHAP utilizes the power

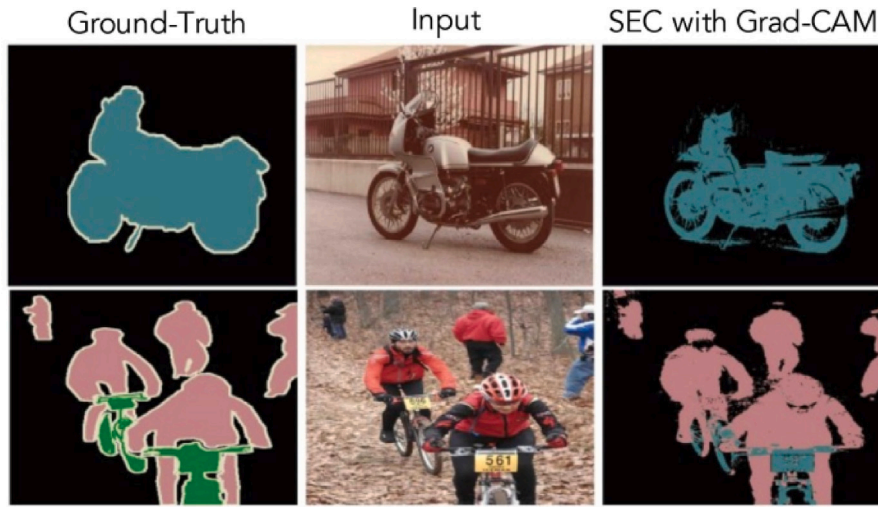


Fig. 8. PASCAL VOC 2012 Segmentation (Kolesnikov & Lampert, 2016) results using Grad-CAM as a location identifier.

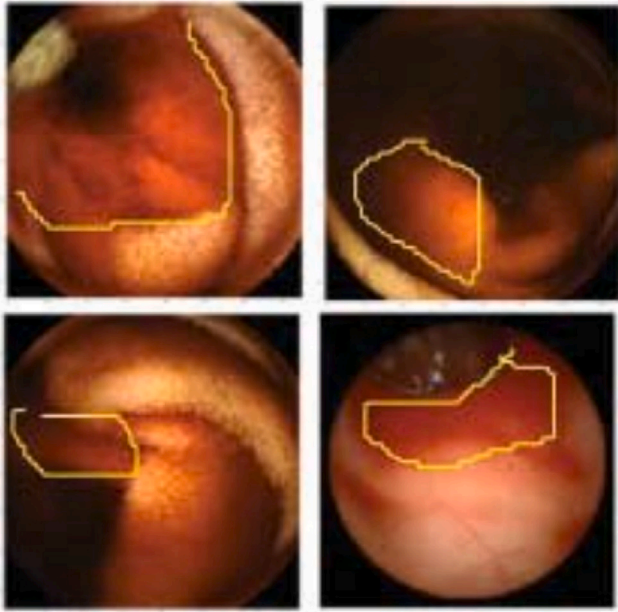


Fig. 9. In-Vivo gastral image classification model explanation using LIME (Malhi et al., 2019).

of deep learning (DL) to produce the Shapley values, the computation time is lightly reduced, but for high resolution images, the computation timing for determining Shapley values can cause issues in practice. Additionally, access to the raw model for backpropagation is required, and thus this process can often be rendered unusable.

7.1.4. LIME

The final prominent example of visual explanations used to assess the behavior of image detection models is LIME. Unlike the previously mentioned examples, LIME does not use back-propagation, and rather generates an explainable surrogate model that it can use traditional techniques on. This technique uses perturbation to modify the input image in various ways, to interpret how the model generates new outputs. These perturbations can include occlusion, shading or morphing, and through making these changes, and analyzing the resulting output, the surrogate functions are formed. The similarity of the perturbed input to the original input is used to monitor the effect that the output of the

perturbed images have on the final explanation (ensuring highly perturbed images will have less impact on the final explanation) (van der Velden et al., 2022). As previously explained, LIME is a locally explainable method, and is highly dependent on the development of a quality surrogate model to provide a quality explanation. In other words, it is not interacting with the inner workings of the black-box model, but rather creates an explainable white-box model that would theoretically produce the same output as the white-box model, and is thus solely dependent on the rules learned by the derived white-box model. The output of the LIME model highly depends on the nature of the surrogate model created, but it ultimately is attempting to identify pixel regions which the model deems important. Through its use of perturbations, these regions can be identified and outlined on the original image, similarly to the previously explained approaches. Malhi et al. used a highlighted boundary to identify the important region of the image for their In-Vivo Gastral image classification model (Malhi et al., 2019), as shown in Fig. 9.

After addressing each of these commonly used visual explanation methods for interpreting the behavior of image detection/classification models, it is clear that there are many different methods that achieve similar outcomes. Each of these visual explanations is ultimately attempting to identify the regions in the original input image that were important for the model's eventual classification, and furthermore, their output overlays the important regions on the original image. As a result, these altered input images, showcasing important regions, can potentially be used to further assess the quality of the explanation.

7.2. Textual explanations

Another form of explanation used for image classification or detection models involves providing textual explanations outlining the important features that constitute the model's decision. These techniques often pair two separate stages, wherein first the feature importance is measured and second the written explanation is generated.

7.2.1. Image captioning (with visual explanations)

As described above, image captioning aims to provide language descriptors for events in an image. In performing this action, it becomes evident what distinct features the model recognizes in the image, and thus one can identify how the model is distinguishing between images. In Vinyals et al.'s 2015 paper (Vinyals et al., 2014), one such example of this is shown, where the concept behind a recurrent neural network (RNN) is revised to pair a convolutional neural network (CNN) and a long-short term memory (LSTM) network. Normally, an RNN operates

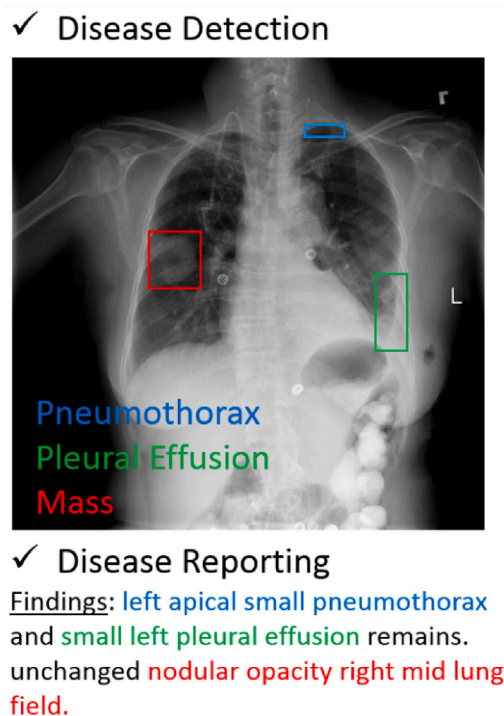


Fig. 10. Automated chest X-ray textual explanation generated in Wang et al.'s TieNet paper (Wang et al., 2018).

with an “encoder” and a “decoder”, where the “encoder” performs some action to obtain important information from the initial input, and the “decoder” then uses the encoder’s output as input, and attempts to generate new information from it. In Vinyals et al.’s paper, the CNN acts as the encoder, and the LSTM acts as the decoder. More specifically, the CNN extracts relevant features from the model, and the LSTM is trained to map those recognized features to a logical description. This logic can be extended to pair any model with a corresponding LSTM model, which can take important features and map them to a logical description. Additionally, this technique can be paired with visual techniques to showcase the results from the first stage (of identifying important model features). This approach was demonstrated by Wang et al. where a saliency mapping technique was first used, and then passed to an attention-encoded text embedding model to generate proper descriptions (Wang et al., 2018). Sample results from this paper can be seen in Fig. 10. The largest hindrance to implementing this method is generating the training data for the LSTM decoder. In Vinyals et al.’s paper, Amazon Mechanical Turk was used to generate descriptions for generic images, although in more nuanced fields, like medicine, these descriptions are far more difficult to generate.

7.2.2. Testing with Concept Activation Vectors (TCAV)

Testing with Concept Activation Vectors (TCAV), presented by Kim et al. (2016), introduces another explainable AI technique wherein human-friendly linear interpretations of neural networks can be determined using a post-hoc method. As a result, this method can be used completely independent of the type of model, and is thus model agnostic.

The basic principle of TCAV states that using user-defined examples of relevant concepts, and random non-relevant concepts, the model can interpret the relevancy of concepts to detected objects. For instance, given the user-defined concept of ‘wheels’, the TCAV approach can measure the degree of which the model uses ‘wheels’ to find if the image contains a bike. The technique uses the underlying math of

concept activation vectors (CAVs), which quantify the sensitivity of such concepts to the trained model’s output. The CAVs learn concepts by analyzing a positive and negative set of images (positive having the concept in them, and negative not having in them).

This means that, in practice, several essential concepts can be defined by a user, and the significance of each of those concepts can be measured, as determined by the model. Kim et al.’s paper also addresses a real-world medical example wherein this technique is used (Kim et al., 2017). Here, doctors identify several concepts that are useful in identifying an ailment. The CAVs are trained using positive and negative sets of images, and finally the relevancy of each of these concepts to the model’s eventual prediction can be realized. The results from this experiment can be seen in Fig. 11, wherein the TCAV scores reflect each of the four trained concepts’ relevancy to the models’ prediction.

This is the first explained method that severely differs from saliency mapping, as it is quantifiably mapping feature importance with a pre-defined knowledge of concepts, as opposed to inadvertently finding relevant concepts in the image through the identification of important sections of the image itself. As a result, this method restricts its explanation to only well defined concepts. This can be beneficial, in that all identified concepts are well defined and understood by the user, but can also be problematic as the user may fail to define unknown yet relevant concepts.

Each of the concepts previously discussed is among the most commonly used for xAI on image detection/classification models. Other approaches exist, such as explanation based approaches (as shown in Fig. 12), yet have a much lower use than visual or textual techniques, and furthermore recycle many of the ideas and approaches already discussed. For instance, Chen et al. introduces a layer to the training process of CNNs that attempts to classify pre-existing groups as a part of the training process. This allows the model to, after the fact, explain itself by giving examples in the testing set to mention image features that look similar. Yet using concepts such as TCAV or saliency mapping, one can already identify important features that the model uses in its inference, and identify similarities in those features after the fact. As a result, this paper will not explore any other xAI concepts related to image detection, and furthermore will focus on the discussed topics when outlining the benefits of the proposed metric.

8. Explainable AI techniques related to Generative AI models

As Generative Artificial Intelligence (GenAI) systems become more sophisticated — capable of generating text, images, audio, and video — the complexity and opacity of their decision-making processes present significant challenges (Adadi & Berrada, 2018). xAI techniques are evolving to help provide insights into how these models generate content, helping users understand and further trust their outputs. Among the benefits of this evolution, integrating xAI with GenAI concepts can improve GenAI model interpretability, enhance user trust, and introduce the ability to detect model biases or unintended outputs, which is especially important in high-stakes applications like health-care, finance, and legal industries. However, the application of xAI to GenAI presents unique challenges, as outlined by Schneider (2024), due to the complexity, scale, and interactive nature of these models. A significant barrier is the lack of access to the internals of commercial GenAI models from corporations like OpenAI and Google, restricting researchers’ ability to apply traditional xAI methods. Additionally, GenAI models are among the largest AI systems, processing vast datasets, having hundreds of billions of parameters (Schneider et al., 2024) and incorporating multiple external tools, making it difficult to trace and interpret how decisions are made. The outputs of GenAI are also inherently complex, involving numerous decisions per artifact rather than a single classification, leading to challenges in explaining aspects such as tone, style, and semantics. Evaluating explanations is particularly difficult, as function-grounded methods rely on benchmarks

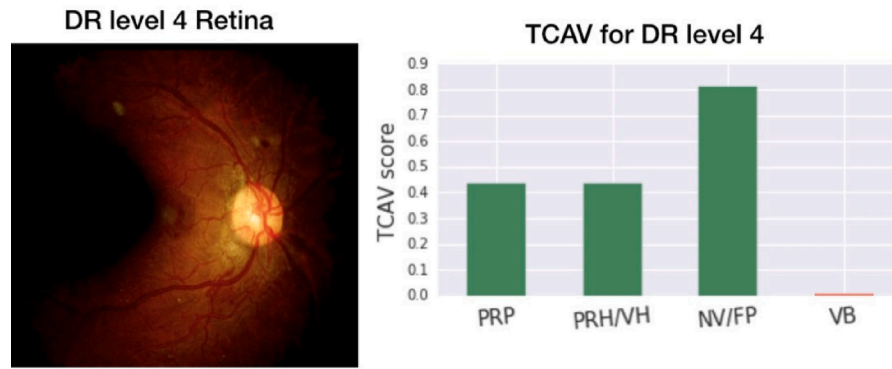


Fig. 11. TCAV scores for a real-world problem of predicting diabetic retinopathy (DR). The graph on the right shows the relevancy of each of the relevant doctor defined concepts (Kim et al., 2017).

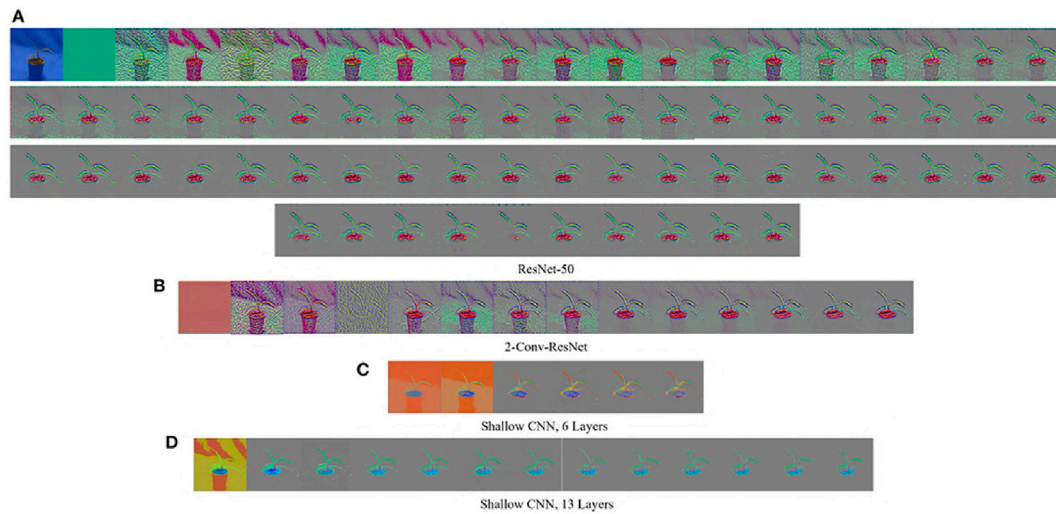


Fig. 12. Guided-Backpropagation results for the Weedling dataset using four varying image detection models: ResNet-50, 2-Conv-ResNet, Shallow CNN with 6 layers and Shallow CNN with 13 layers (Mostafa et al., 2022).

that may not exist for certain GenAI tasks. Further complicating the explainability of GenAI are its diverse user base and ethical risks. Unlike other aforementioned AI models, which are often tailored to specific tasks and user groups, GenAI is widely used across different demographics, necessitating explanations that are comprehensible to all. Ethical concerns also arise as GenAI models can produce biased or harmful content, and even their self-explanations may inadvertently include offensive elements (OpenAI et al., 2023). Additionally, GenAI suffers from technical shortcomings like hallucinations and limited reasoning capabilities, meaning that its explanations may also be unreliable. These challenges highlight the ongoing need for novel xAI approaches tailored specifically to the intricacies of GenAI, ensuring transparency and trust while accounting for its scale, interactivity, and ethical considerations.

With this in mind, there have been some prominent explainable AI approaches for interpreting generative models, including incorporating Layer-wise Relevance Propagation (LRP) (Bach et al., 2015), Local Interpretable Model-Agnostic Explanations (LIME) (Ribeiro et al., 2016), and other explanation methods crafted explicitly for Generative Adversarial Networks (GANs). LRP is a popular explainability technique that enables attribution of relevance to input features of deep neural networks (Montavon et al., 2017). It decomposes the predictions of a model into contributions of individual features, making it easier to understand which features influence the model's output. This technique has been applied to various AI models, including image classification, text classification, and generative models such as Variational Autoencoders (VAEs) (Nguyen et al., 2016). LIME is another widely-used

model-agnostic approach for explanation, providing local explainability around a prediction by fitting simpler models, such as linear classifiers or regressions, to the complex model being explained (Ribeiro et al., 2016). This method has been extended to generative AI models, including GANs, allowing users to understand how different input features impact the generation process (Sangkloy et al., 2016). In the context of GANs, various explanation techniques have been proposed to enhance transparency and explainability. For instance, the InfoGAN model (Chen et al., 2016) extends the original GAN framework by maximizing the mutual information between a subset of latent codes and the output. This approach encourages the generative process to capture explainable semantic features, aiding in the understanding of the internal mechanisms of GANs. Furthermore, the Disentangled Representation GAN (DR-GAN) (Kim & Mnih, 2018) is designed to learn entirely disentangled and explainable representations by developing a new regularization term containing mutual information, allowing users to manipulate specific aspects of generated samples.

GANs have emerged as a popular and powerful tool for generating synthetic data, such as images, audio, and text, by learning from a given dataset. However, the interpretability of these models has been a challenging area of research. GAN-based explanation techniques have been developed to provide insights into the inner workings of these models and understand the decision-making processes behind the generated outputs. One such technique is GAN Dissection (Bau et al., 2018), a method to visualize and manipulate the GANs internal representations. In this approach, the researchers identify the neurons activated for

specific features, for example, doors, windows, trees, and manipulate or remove these features to analyze the role of each neuron in generating the output. Another technique, called DragGAN (Pan et al., 2023), is a novel method that allows users to interactively “drag” specific points in GAN-generated images to desired target positions, enabling precise control over attributes such as pose, shape, expression, and layout across various object categories. This is achieved through two main components: a feature-based motion supervision that guides handle points toward target positions, and a point tracking approach leveraging discriminative generator features to continuously localize handle points during manipulation. While GAN-based explanation techniques have significantly contributed to improving the interpretability of Generative AI models, there is still room for more research, innovation, and collaboration, given the complexity of these models and the growing need for explainable AI in various applications and industries.

8.1. Layer-wise Relevance Propagation (LRP)

As has been touched on, Layer-wise Relevance Propagation is an explainable AI technique that has gained traction due to its ability to interpret deep learning models, particularly in the field of Generative AI (Bach et al., 2015). In essence, LRP aims to attribute relevance scores to neurons within each layer in the network by decomposing the output prediction into contributions from input features. This approach makes it possible to visualize the importance of individual input features, thus gaining insights into the decision-making process of neural networks (Montavon et al., 2017).

While Layer-wise Relevance Propagation (LRP) has been successfully applied to enhance the explainability of deep learning models in various domains, its specific application to Generative Adversarial Networks (GANs) for elucidating the roles of latent and observed variables is an emerging area of research. One recent work though, written by Wang et al. (2025), proposes an explainable framework that equips a GANs’ latent space with semantic classification capabilities without additional training. This framework facilitates the understanding of how latent variables generate semantically meaningful images, offering insights into the data distribution learned by the model.

Additionally, it has been applied to help clarify a GANs models’ predictions in medical settings, like in Ahmed et al.’s work to help segment polyps and instruments (Ahmed & Ali, 2021). More specifically, the LRP process was utilized to identify which input pixels significantly influenced the segmentation outcomes, thereby providing insights into the decision-making process of the GANs. This integration of LRP with GANs enabled the generation of visual explanations, highlighting the regions in input images that were most relevant to the model’s predictions.

Overall, LRP provides a tried explainable framework for interpreting Generative AI models, helping shed light on the complex inner workings of these deep learning models and improve the trust and transparency in AI-driven systems.

8.2. LIME for generative AI

As explained previously, LIME is a widely-used explainable AI technique that has applications across many different types of AI models. In Generative AI models, LIME can be employed in various tasks, such as image synthesis and natural language processing, to unravel the factors that contribute to the model’s predictions. Several researchers have demonstrated the efficacy of applying LIME to interpret Generative AI models in the image synthesis domain. For example, in the context of natural language processing, LIME provides insights into the significance of individual input words or phrases in affecting the predictions of a generative language model (Alvarez-Melis & Jaakkola, 2017). Benchama and Zebbara (2024) have also recently published an innovative system that combines GANs, Multi-Scale Convolutional Neural Networks (MSCNNs), Bidirectional Long Short-Term Memory (BiLSTM)

networks, and LIME to enhance intrusion detection in network security. This is particularly important because, without such explanations, it can be difficult to trust the model’s predictions, especially in high-stakes applications like cybersecurity. The GAN component generates realistic network traffic data, encompassing both normal and attack patterns, which is then processed by the MSCNN-BiLSTM architecture. The MSCNN extracts features at multiple scales, while the BiLSTM captures temporal dependencies within the traffic sequences. Finally, LIME provides interpretability to the process, making the decision-making process of their hybrid model more transparent, enabling network administrators and security experts to understand the reasoning behind classifications of network traffic. LIME acts here by approximating the decision boundaries of the model locally, meaning it provides insights into the features that influence predictions for individual instances of network traffic. In the context of intrusion detection, LIME helps to explain which specific features of the network traffic — such as packet size, protocol type, or timestamp — are most influential in classifying the traffic as normal or as an attack (Benchama & Zebbara, 2024).

As with other xAI techniques, future developments for LIME and its continued application to Generative AI models, will be a prominent area of research for years to come, as GANs continue to be heavily used and require more scrutiny around understanding their actions when used in various high-risk domains.

8.3. Overview of all discussed xAI types

See Table 3.

9. Explainable AI technique applications in various industries

Especially in recent years, explainable AI (xAI) techniques have significantly impacted various industries, including finance, healthcare, retail, manufacturing, and autonomous systems. In the finance sector, xAI enables financial institutions to understand and explain automated decision-making, gaining trust in credit scoring and risk management models, providing more insight into model behaviors and input impacts. In healthcare, xAI techniques like LRP and Grad-CAM provide insights into how deep learning models predict diseases, improving the transparency of algorithms for medical professionals and patients, especially when handling images. These techniques have been used for diagnosing brain disorders, detecting breast cancer, and predicting potential outbreaks of infectious diseases. The retail industry significantly benefits from xAI in demand forecasting, personalized marketing, and customer analytics. Using techniques like feature importance and partial dependence plots, retailers can better understand customer preferences and make informed decisions about inventory and pricing. Manufacturing industries leverage xAI techniques like Layer-wise Relevance Propagation to detect patterns in sensor data and improve predictive maintenance and quality control. For example, xAI can help identify root causes for equipment failure and prioritize maintenance tasks. The rise of autonomous systems also demands a higher level of transparency and trust. Explainable AI plays a crucial role in developing safe, reliable, and accountable self-driving vehicles and drone systems. Techniques like visual analytics, counterfactual explanations, and model-agnostic methods provide insights into the decision-making process of these complex systems and help reduce accidents and errors.

9.1. Healthcare

Explainable Artificial Intelligence (xAI) has made significant strides in the healthcare industry, offering substantial benefits in terms of improving model transparency, trust, and interpretability. In an environment where the stakes are high, such as healthcare, the ability to understand and explain AI-driven decisions is critical for ensuring both clinical effectiveness and patient safety. While AI systems, including

Table 3

Overview of all discussed xAI types.

Explainable AI technique	AI model type implementations	Local or global explainability	Post-Hoc or Ante-Hoc
Attention-based Mechanisms	- Supervised Learning Models - Reinforcement Learning Models	Global	Ante-Hoc
Capsule Networks (CapsNets)	- Supervised Learning Models - Unsupervised Learning Models - Reinforcement Learning Models	Global	Ante-Hoc
Neural Symbolic Networks (NSNs)	- Supervised Learning Models - Unsupervised Learning Models - Reinforcement Learning Models	Global	Ante-Hoc
Self-Explaining Neural Networks (SENNs)	- Supervised Learning Models - Unsupervised Learning Models - Reinforcement Learning Models	Global	Ante-Hoc
Quantized Self-Explaining Neural Networks (QSENNs)	- Supervised Learning Models - Unsupervised Learning Models - Reinforcement Learning Models	Global	Ante-Hoc
LIME	- Supervised Learning Models - Unsupervised Learning Models - Reinforcement Learning Models - Generative AI Models	Local	Post-Hoc
Shapley Values	- Supervised Learning Models - Reinforcement Learning Models (in some capacity) - Generative AI Models (in some capacity)	Local	Post-Hoc
Partial Dependence Plots (PDPs)	- Supervised Learning Models - Unsupervised Learning Models	Local	Post-Hoc
Individual Conditional Expectation (ICE) Plots	- Supervised Learning Models - Unsupervised Learning Models	Local	Post-Hoc
Counterfactual Explanations	Supervised Learning Models	Local	Post-Hoc
Anchors	Supervised Learning Models	Local	Post-Hoc
Model Distillation	Supervised Learning Models	Local	Post-Hoc
t-SNE	- Supervised Learning Models - Unsupervised Learning Models	Global	Post-Hoc
Bayesian Rule Lists	Supervised Learning Models	Local	Ante-Hoc
Saliency Mapping	- Supervised Learning Models - Reinforcement Learning Models - Generative AI Models	Local	Post-Hoc
Local-Linear Models	- Supervised Learning Models - Reinforcement Learning Models	Local	Post-Hoc
Guided Backpropagation	Supervised Learning Models	Local	Post-Hoc
Grad-CAM	Supervised Learning Models	Local	Post-Hoc
Deep-SHAP	- Supervised Learning Models - Generative AI Models	Local	Post-Hoc
Testing with Concept Activation Vectors (TCAV)	Supervised Learning Models	Global	Ante-Hoc

deep learning models, have demonstrated exceptional performance in tasks such as medical image analysis (Esteva et al., 2017), disease prediction, and drug discovery (Vamathevan et al., 2019), their “black-box” nature often limits their adoption in clinical settings. This, of course, is where xAI plays a pivotal role in bridging the gap between model accuracy and clinical trust.

One of the key areas where xAI has been applied is in the interpretation of medical imaging. Models such as convolutional neural networks (CNNs) have been widely used to classify medical images, for instance, in detecting tumors in radiology scans. However, due to their opacity, clinicians may hesitate to rely on the results provided by these models. Tools like LIME (Ribeiro et al., 2016) and SHAP (Lundberg & Lee, 2017) have been leveraged to provide explanations of CNN model predictions, allowing clinicians to understand which parts of the image were most influential in the model’s decision.

Other xAI techniques have been developed explicitly for use in the medical field, such as RETAIN (Recurrent Time Attention), which is specifically designed for medical time-series data (Choi et al., 2016). RETAIN employs an attention mechanism to highlight the most relevant

parts of a patient’s medical history, aiding in predictive tasks like patient readmission or disease progression. Unlike general-purpose xAI techniques like LIME or SHAP, RETAIN offers local explainability tailored to sequential medical data, enhancing interpretability in health-care settings and improving trust in AI-driven predictions. However, as an ante-hoc process (since it incorporates interpretability directly into the model’s architecture), it cannot be used for general model applications, and a RETAIN-based model must be integrated for its explicit use. Additionally, while RETAIN is highly specialized for time-series data, it may not be as flexible or generalizable to other types of data or domains outside healthcare.

Moreover, xAI techniques are also valuable in predictive healthcare, such as predicting patient outcomes or diagnosing diseases. Many machine learning models, including those used for risk prediction, are trained on vast amounts of patient data, making it challenging for clinicians to interpret the rationale behind the predictions. Using xAI tools, it becomes possible to extract meaningful insights into the decision-making process. For instance, Caruana et al. (2015) used rule-based models to predict sepsis, and xAI methods provided interpretable

explanations that allowed healthcare providers to better understand the reasons behind these predictions and take necessary preventive actions. As explained above as well, [Ahmed and Ali \(2021\)](#) propose a method for automating medical image segmentation using GANs combined with Layer-wise Relevance Propagation LRP, focusing on segmenting polyps and instruments in endoscopy images, addressing the need for accurate and explainable segmentation in medical diagnostics. This approach not only improves segmentation performance but also provides clinicians with insights into the decision-making process of the AI system, fostering trust and facilitating the validation of AI-driven analyses in medical imaging.

In drug discovery, xAI also proves valuable by providing interpretability in complex models that predict molecule interactions and potential drug efficacy. [Jiménez-Luna et al. \(2020\)](#) highlighted various techniques in how xAI techniques could be used to interpret the model's learning from chemical data, helping researchers to make more informed decisions about drug candidates. This interpretability can also reduce biases in the model, ensure fairness, and lead to more ethical decision-making in drug development processes.

Despite its potential, there are challenges to implementing xAI in healthcare. One of the main issues is the need for domain expertise in interpreting the explanations provided by xAI tools. Moreover, there is an ongoing challenge in ensuring that xAI models are not only accurate but also generalizable across diverse populations and clinical scenarios ([Gilpin et al., 2018](#)). Finally, integrating xAI into clinical workflows requires overcoming regulatory hurdles, ensuring that explanations provided by models are understandable to non-expert stakeholders, including patients.

As shown, xAI has the potential to significantly transform the healthcare industry by making AI systems more transparent, trustworthy, and clinically actionable. By enabling clinicians to understand the reasoning behind AI predictions, xAI fosters improved decision-making, enhances patient safety, and ultimately supports better healthcare outcomes.

9.2. Retail

Explainable Artificial Intelligence (xAI) has gained significant traction in the retail industry, where data-driven decisions and customer insights play a pivotal role in business strategy. The application of xAI techniques has proven essential in making AI-driven processes transparent and understandable for retail professionals, enabling better decision-making and improving customer satisfaction.

One of the primary areas where xAI is being implemented in retail is in personalized recommendation systems. Retailers use recommendation algorithms to suggest products to customers based on their past behaviors, preferences, and purchasing patterns. However, these systems can sometimes be opaque, making it difficult for retailers to understand why certain products are recommended. By employing xAI techniques such as LIME ([Ribeiro et al., 2016](#)) and SHAP ([Lundberg & Lee, 2017](#)), companies can interpret the reasoning behind the recommendations. These methods provide valuable insights into which features of the customer's profile — such as purchase history or browsing behavior — contribute most to a specific recommendation. This transparency helps to increase customer trust in automated systems, ensuring that suggestions are aligned with consumer needs and preferences. Moreover, xAI plays a critical role in targeted marketing campaigns. Using xAI, retail marketers can understand the rationale behind AI-generated consumer segments and recommendations, enabling the development of more effective strategies to improve customer conversion and retention ([Njoku et al., 2024](#)).

In inventory management, xAI is also being applied to predict demand and optimize stock levels. Machine learning models are used to forecast product demand based on factors like seasonal trends, customer purchasing behavior, and external factors such as market conditions. With the use of xAI, retailers can gain insight into which features most significantly influence sales predictions, helping inventory

managers make better decisions about stock levels, and business leaders make better decisions regarding marketing and sales tactics ([Chen et al., 2024](#)).

Moreover, xAI plays a vital role in pricing optimization. Retailers use dynamic pricing strategies, adjusting prices in real-time based on demand, competitor pricing, and customer preferences. By making the price optimization model interpretable, xAI allows retailers to understand the key factors driving price adjustments and identify potential biases in pricing algorithms, ensuring fairness and transparency, especially in the data-driven world of e-commerce ([Sarkar, 2024](#)).

The retail industry also benefits from xAI in customer service automation, where AI-powered chatbots and virtual assistants help handle customer queries. Explainable AI models help assess and improve the chatbot's decision-making process by providing insights into how the model arrives at responses. This not only enhances customer satisfaction but also enables businesses to quickly identify areas where the chatbot can be improved to better serve customer needs ([Kovari, 2025](#)). Supply chain optimization is another significant application of xAI in this area, where AI-driven tools can predict fluctuations in demand, enabling retailers to better allocate resources, schedule deliveries, and maintain appropriate inventory levels. By providing more transparent predictions, xAI techniques can help establish trust between retailers, suppliers, and AI systems [4].

Despite the promising applications, there are several challenges to the widespread adoption of xAI in the retail sector. One major challenge is the complexity of some AI models used in retail applications, which can make them inherently difficult to explain. Additionally, there is a need for standardized evaluation metrics to assess the effectiveness of xAI techniques across different retail applications. Retailers must also navigate data privacy regulations and ensure that the explanations provided by AI models comply with industry standards and consumer expectations ([Adanyin, 2024](#)).

As the retail industry continues to embrace AI-driven technologies, the development of more efficient and understandable xAI models will be crucial to improving both operational efficiency and customer experience.

9.3. Finance

The finance industry has increasingly adopted artificial intelligence models to enhance decision-making processes, from credit scoring and fraud detection to algorithmic trading. However, the “black-box” nature of various models leads to a high risk for profound financial and regulatory consequences, which hinder the adoption of these powerful strategies. xAI offers an essential solution for maintaining trust, ensuring fairness, and complying with regulatory requirements. This leads to many pertinent financial implementations of xAI. One such key application of xAI in finance is credit scoring, where AI models are used to assess the creditworthiness of individuals and organizations. Traditional credit scoring methods are often based on linear models, which are easier to interpret but may fail to capture complex patterns. In contrast, AI-based credit scoring models, such as those utilizing decision trees or deep learning, are more accurate but are often opaque, where xAI can help the continued adoption of these models. Nallakaruppan et al. recently proposed an Explainable AI (xAI) framework using Random Forest integrated with LIME and SHAP for loan rejection and acceptance predictions, addressing the need for this transparent decision-making in banking systems, particularly post-pandemic, with the framework offering high accuracy and providing explanations through local and global surrogates ([Nallakaruppan et al., 2024](#)).

Another critical application of xAI in finance is fraud detection. AI models are widely used to detect fraudulent transactions by analyzing transaction data in real-time. However, as these models grow in complexity, understanding the reasoning behind flagged transactions

becomes challenging. xAI methods, such as counterfactual explanations, can help explain exactly why a particular transaction was flagged as fraudulent by highlighting the specific features that deviated from the norm (Caruana et al., 2015). This transparency not only improves trust among stakeholders but also allows financial institutions to better communicate and justify decisions to customers and regulators. Zhou et al. also address this requirement for explainability in financial fraud detection by providing local and global explanations to provide stakeholders with a user-centered explainable framework for financial fraud detection models (Zhou et al., 2023). The approach combines an ensemble model with Shapley values to ensure both high accuracy and explainability, including local explanations (for individual predictions) and global explanations (for the overall model), addressing the needs of key stakeholders such as regulators, auditors, analysts, and investors. This framework ensures that financial fraud detection is not only accurate but also transparent, making it more reliable and understandable for stakeholders. This paper highlights an important aspect of utilizing xAI in practice, as explained by Zhou et al. in their paper (Zhou et al., 2023), while the explanations provide insight into the key features influencing model predictions, they do not directly identify the causal factors driving the outcome. This gap means that while xAI can improve model transparency and user trust, it may still fall short in providing causal explanations that could be crucial for understanding the root causes of decisions, especially in domains like fraud detection, where pinpointing specific causes is important for prevention and risk management.

A final prominent example of using xAI in the financial space is algorithmic trading, with models being used to predict market trends, optimize trading strategies, and execute trades automatically. The lack of interpretability in these models, especially in high-frequency trading, can raise concerns regarding the accountability of AI-driven market actions. By applying xAI techniques, traders may gain insights into how different market features (e.g., stock price, trading volume) affect the predictions of trading models. This transparency is crucial for mitigating the risks of unintended consequences in volatile markets. However, in the world of algorithmic trading, researchers have outlined the necessity for a multi-faceted evaluation approach to ensure the practical utility of trading models (Jabbar & Jalil, 2024), which leads to xAI approaches that are heavily model agnostic and Post-Hoc in nature, such as LIME or SHAP, but one must be careful about the local nature of these models when applying them in a stock-market setting, due to the ever evolving trends of the underlying data.

As AI continues to evolve in the finance sector, the integration of explainability will be key to its successful and ethical deployment.

9.4. Manufacturing

Finally, looking at the manufacturing industry, explainable AI (xAI) techniques are increasingly vital, offering transparency and interpretability to complex AI models used for predictive maintenance, supply chain management, quality control, and process optimization.

As mentioned, predictive maintenance is a critical application where AI models are used to predict potential equipment failures before they occur. This helps minimize downtime, extend equipment life, and reduce maintenance costs. By providing interpretable insights, these xAI methods can allow maintenance teams to prioritize repairs with greater confidence. In their paper titled “Explainable Predictive Maintenance”, Pashami et al. address the importance of xAI in predictive maintenance approaches, highlighting the diverse explanation needs of various stakeholders in industrial settings through four industrial case studies: commercial vehicles, metro trains, steel plants, and wind farms (Pashami et al., 2023). Dereci et al. showcases the explicit integration of LIME with one of these systems as well, showcasing its practicality and usefulness when understanding reasoning behind the binary outcome of the model (Dereci & Tuzkaya, 2024).

AI has also been applied to optimize supply chain management, where models are commonly used to predict demand, optimize inventory, and improve logistics. However, the decisions made by these models can have significant economic implications, and stakeholders must understand the rationale behind these decisions. By providing these insights, xAI ensures that decision-makers, such as supply chain managers and logistics coordinators, can make more informed, data-driven decisions with a better understanding of the model’s logic. Recent research on neural symbolic network approaches (as described in Section 2.3 as an Ante-Hoc approach) within supply chains have focused on enhancing explainability by combining neural networks with logic-based reasoning. A systematic review highlighted the growing interest in these approaches but pointed out gaps in areas like performance evaluation and sustainability (Kosasih et al., 2024). The paper proposes five pillars for advancing neural symbolic AI in supply chains, alongside use cases to help address key supply chain challenges and promote adoption of these technologies. Post-hoc approaches can also be utilized for supply-chain management, as shown in W. De Bock et al.’s paper wherein they highlight the specific application of techniques like LIME, SHAP or PDP to Operation Research type problems, many of which have direct applications to supply chain management use-cases (Bock et al., 2024).

In quality control, AI models are often used to detect defects in products during manufacturing, ensuring that only products that meet quality standards reach customers. Deep learning models, particularly one in the computer vision subset of models like convolutional neural networks (CNNs), are commonly used to analyze images of products and identify defects. As described before, this means common xAI models used for quality control might include Guided Backpropagation, Grad-CAM, Deep SHAP, LIME or TCAV. As outlined by Müller et al. these approaches are crucial in understanding the reasoning behind quality defects within manufacturing firms and several implementations can have major impacts within a production line in regards to time saved when understanding the root cause of these defects (Müller et al., 2022).

Across each of these examples, the power of xAI is made clear in facilitating better decision-making, enhancing operational efficiency, and supporting the broader adoption of AI technologies in the manufacturing sector.

10. Evaluating xAI

Evaluating xAI techniques requires structured frameworks to assess explanation quality, reliability, and utility. This section briefly provides a compact taxonomy of evaluation metrics and methods, enabling practitioners to judge explanations systematically.

10.1. Evaluation metrics taxonomy

Key dimensions include: (1) *Faithfulness/fidelity*—how accurately explanations reflect actual model behavior, measured by removing important features and observing output changes or comparing explanation-based predictions to actual predictions (Alvarez-Melis & Jaakkola, 2017); (2) *Stability*—consistency of explanations for similar inputs; (3) *Robustness*—performance under adversarial perturbations, input noise, or model variations; (4) *Simulatability*—extent to which humans can predict model output from explanations; and (5) *Comprehensibility*—ease of human understanding based on length, complexity, and terminology.

10.2. Local versus global fidelity

Local fidelity measures how well explanations capture model behavior for specific instances (e.g., LIME (Ribeiro et al., 2016), SHAP (Lundberg & Lee, 2017)), while *global fidelity* assesses overall model behavior across datasets (e.g., feature importance, partial dependence plots (Friedman, 2001)). The choice depends on application needs: local for specific predictions, global for overall model understanding.

10.3. Robustness testing

Essential validation methods include: (1) *input perturbation tests*—observing explanation consistency under small input variations; (2) *sanity checks*—verifying explanations reflect model behavior (e.g., randomizing model weights to test if explanations change appropriately); (3) *sensitivity analysis*—examining how explanations change with model parameter variations; and (4) *consistency checks*—comparing explanations across methods or multiple runs.

10.4. Human-centered evaluation

Two study types are critical (Doshi-Velez & Kim, 2017): (1) *human-grounded studies*—testing explanation understandability in controlled settings with simplified tasks; and (2) *application-grounded studies*—evaluating explanations in realistic deployment contexts, measuring impact on actual user tasks and decision-making. Both are essential for comprehensive evaluation.

10.5. Evaluation practices by technique

Established practices exist for major techniques: LIME (Section 3.1) emphasizes local fidelity and stability across random seeds; SHAP (Section 3.2) validates faithfulness to Shapley axioms (efficiency, symmetry); Partial Dependence Plots (Section 4.1) assess accurate feature effect representation; attention mechanisms (Section 2.3) require faithfulness checks and counterfactual tests given faithfulness concerns; and Grad-CAM (Section 6.1) uses quantitative metrics (localization accuracy, deletion/insertion tests) and human studies.

10.6. Industry frameworks

Industry evaluation emphasizes *actionability* (enabling meaningful user actions), *trust metrics* (improving user confidence), *compliance* (meeting regulatory requirements), and *efficiency* (production-ready performance). Standardized frameworks remain an active area of development.

11. Opportunities and future challenges for explainable AI

After analyzing many of the preferred xAI methods used today, we can identify some of the opportunities and challenges that arise from using xAI post-hoc approaches. Firstly, the level at which interpretability is required in a project will still highly influence the choice of model in many real-world scenarios today, even with the introduction of xAI (Rudin, 2018). The level of transparency that simple models such as regression and basic decision-trees have, is unmatched by any current post-hoc xAI approach, and in situations where a need for explainability heavily outweighs the need for accuracy, transparent models will still be chosen. Additionally, many individuals and AI experts are still unaware of these approaches, and generating explanations that are digestible for those with non-technical expertise, is still an open issue in the field (Wachter, Mittelstadt, & Floridi, 2017). Yet not all issues with xAI relate to its pending acceptance from the public. Generally, xAI aims to provide complex models with a measure of explainability, and still there is no given metric to gauge interpretability of one technique when compared to others. Without consistent metrics to gauge explainability, it is impossible to interpret which xAI techniques outperform others. Recent attempts at developing a metric have been studied by individuals such as Hoffman et al. (2018) and Mohseni et al. (2021), but neither provided a general enough standard by which to compare all xAI techniques (Arrieta et al., 2020). As a result, though new xAI techniques may be developed, it is extremely difficult to measure their impact on the community without a general metric for explanation trustworthiness and reliability. As shown, the field of xAI is evidently ever changing with the endless development of increasingly

complex ML models, and its importance is paramount to the adoption of complex AI models in real-life applications. As a result, unlocking the power of AI is intricately tied to identifying methods to interpret model behavior. In upcoming years, it should be expected that the field of xAI will continue to grow, and breakthroughs in xAI techniques to make deep learning models more transparent or in usable metrics to compare general xAI approaches against one another, could lead to a significantly more powerful and cognitive tomorrow. Yet, as shown, a significant gap in the literature today pertains to a lack of generalizable metrics to compare xAI techniques.

11.1. Ethical considerations in explainable AI

As xAI systems become increasingly deployed in high-stakes applications, ethical considerations have emerged as a critical aspect of responsible AI development and deployment. Understanding and addressing ethical challenges is essential for ensuring that xAI systems not only provide transparency but also promote fairness, accountability, and trustworthiness (Balasubramaniam et al., 2023).

One of the primary ethical concerns in xAI is the potential for bias in explanations. While xAI techniques can help reveal model biases by highlighting which features influence predictions, they can also perpetuate or amplify existing biases if not carefully designed and validated. For instance, if a model exhibits gender or racial bias in its training data, explanation methods like SHAP or LIME may highlight features that reflect these biases, potentially reinforcing discriminatory patterns in decision-making (Lundberg & Lee, 2017). Conversely, xAI can serve as a valuable tool for bias detection and mitigation by making model decision-making processes transparent, allowing practitioners to identify and address unfair patterns before deployment. The challenge lies in ensuring that explanations themselves do not introduce new biases or mislead users about the fairness of model decisions.

Fairness considerations extend beyond bias detection to encompass how explanations are generated and presented. Different stakeholders may require different types of explanations, and ensuring that explanation methods provide fair and equitable access to understanding model decisions is crucial. For example, in credit scoring applications, xAI explanations should enable all applicants to understand why their application was accepted or rejected, regardless of their technical background or demographic characteristics. This requires developing explanation methods that are not only accurate but also accessible and comprehensible across diverse user groups.

Privacy concerns represent another significant ethical challenge in xAI. Some explanation methods, particularly those that require access to training data or model internals, may inadvertently reveal sensitive information about individuals in the dataset. For instance, model inversion attacks could potentially reconstruct training data from model explanations, raising privacy concerns especially in health-care or financial applications where data confidentiality is paramount. Additionally, explanation methods that highlight specific features may reveal information about protected attributes or sensitive personal characteristics, potentially violating privacy regulations such as GDPR or CCPA (Balasubramaniam et al., 2023).

Accountability and responsibility in xAI systems are also critical ethical considerations. When an AI system makes a decision based on an explanation, determining who is responsible for the outcome—the model developer, the explanation method designer, the end-user, or the system itself—becomes complex (Brand & Nannini, 2023). This is particularly important in high-stakes applications where incorrect decisions can have serious consequences. Clear frameworks for accountability are needed to ensure that responsibility is appropriately assigned when xAI systems fail or produce harmful outcomes.

The potential for explanation manipulation or adversarial attacks on explanations represents an emerging ethical concern. Just as adversarial examples can fool machine learning models, adversarial attacks on explanations could be used to generate misleading or false explanations

that appear trustworthy but do not accurately represent the model's decision-making process. This could be exploited to hide model biases, justify incorrect decisions, or manipulate user trust. Developing robust explanation methods that are resistant to such attacks is an important area for future research.

Finally, ethical guidelines and frameworks for responsible xAI deployment are still evolving (Shulner-Tal & Sheidin, 2025). While general AI ethics principles apply to xAI systems, specific guidelines for explainability — such as what constitutes an adequate explanation, when explanations are required, and how to balance transparency with other considerations like privacy or security — are still being developed by regulatory bodies and professional organizations. As the field matures, establishing clear ethical frameworks will be essential for ensuring that xAI systems are deployed responsibly and benefit society while minimizing potential harms.

12. Conclusion

This comprehensive survey has provided a structured overview of Explainable AI (xAI) techniques organized by machine learning paradigms, addressing a significant gap in existing survey literature. By systematically categorizing xAI methods according to their application across supervised learning, unsupervised learning, reinforcement learning, computer vision, and generative AI models, this work offers practitioners a clear framework for selecting appropriate explanation methods based on their specific ML model types. The detailed exploration of foundational techniques — from LIME and Shapley Values to specialized methods for reinforcement learning and GANs — demonstrates the critical importance of explainability in making AI systems more trustworthy, accountable, and suitable for deployment in high-stakes environments.

The comprehensive coverage of industry applications across healthcare, finance, retail, and manufacturing sectors illustrates the practical significance and real-world impact of xAI implementations. These applications reveal that explainability is not merely a theoretical concern but a practical necessity for regulatory compliance, stakeholder trust, and ethical AI deployment. The identification of key challenges — particularly the lack of standardized evaluation metrics, the persistent interpretability-accuracy trade-off, and the need for explanations digestible to non-technical stakeholders — highlights critical areas requiring continued research and development.

12.1. Benefits and implications

This survey provides several key benefits for different audiences. For researchers, it offers a structured reference that identifies gaps in current literature, particularly the need for generalizable evaluation metrics and standardized frameworks for comparing xAI techniques. For practitioners and AI developers, the paradigm-specific organization enables quick identification of relevant explanation methods, supporting more effective implementation of explainability in production systems. For industry professionals in high-stakes domains, the comprehensive coverage of applications demonstrates how xAI can enhance transparency, regulatory compliance, and stakeholder trust in AI-driven decision-making systems.

The implications of this work extend beyond serving as a reference document. By organizing xAI techniques according to ML paradigms, this survey contributes to the advancement of the field by providing a structured framework that can guide future research directions. The identification of critical gaps — such as the lack of standardized evaluation metrics — can inform funding priorities and research agendas. Additionally, the comprehensive taxonomy supports the development of more systematic approaches to xAI implementation, potentially accelerating the adoption of explainable AI in industry settings.

12.2. Future research directions

Several promising directions for future research emerge from this comprehensive survey. First, the development of standardized evaluation metrics and frameworks for comparing xAI techniques represents a critical priority. Current research efforts by Hoffman et al. (2018) and Mohseni et al. (2021) have made initial progress, but a generalizable standard that can be applied across all xAI techniques remains an open challenge. Such metrics would enable more systematic evaluation of explanation quality, faithfulness, and utility across different contexts and applications.

Second, the field requires significant advances in scalable xAI approaches for large-scale and high-dimensional applications. As AI models continue to grow in complexity and size, developing computationally efficient explanation methods that can handle millions of samples and thousands of features becomes increasingly important. Research into approximation algorithms, distributed computing frameworks, and intelligent sampling strategies will be essential for making xAI practical for real-world industrial applications.

Third, emerging areas such as xAI for edge computing and resource-constrained environments present new challenges and opportunities. As AI systems are deployed on mobile devices, IoT sensors, and embedded systems, there is a growing need for lightweight explanation methods that can operate within strict computational and memory constraints. Similarly, real-time and interactive explanation systems that can provide on-demand explanations during model inference represent an important direction for enhancing user experience and trust.

Fourth, multimodal xAI approaches that can explain models processing multiple data types (text, images, audio, video) simultaneously are becoming increasingly relevant as multimodal AI systems gain prominence. The integration of explanation methods across different modalities presents unique challenges in maintaining consistency and comprehensibility while handling the complexity of multimodal interactions.

Fifth, the explainability of large language models (LLMs) and foundation models represents a critical frontier. As these models become ubiquitous in various applications, understanding their decision-making processes, identifying potential biases, and ensuring their outputs are trustworthy becomes paramount. Research into prompt-based explanations, attention visualization for transformers, and methods for explaining generated content will be essential for responsible deployment of these powerful models.

Sixth, causal explanation methods that go beyond correlation to identify true causal relationships in model decisions represent a promising direction for enhancing explanation quality and utility. Such methods could provide more actionable insights for practitioners and enable better understanding of model behavior under different conditions.

Finally, the development of human-AI collaborative explanation systems that can adapt explanations based on user expertise, context, and needs will be crucial for making xAI more accessible and useful across diverse stakeholder groups. These systems could dynamically adjust explanation complexity, format, and detail level to match the requirements of different users, from technical developers to end-users with limited AI expertise.

As the field of xAI continues to evolve alongside advances in AI and machine learning, addressing these research directions will be essential for realizing the full potential of explainable AI in making AI systems more transparent, trustworthy, and suitable for deployment across diverse applications and industries.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Stephen Andrew Gadsden reports financial support was provided by Natural Sciences and Engineering Research Council of Canada. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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